notebook1

The purpose of this notebook is to document the development of the analysis. The final analysis will be run through a series of scripts, but I will develop those scripts interactively here and document my decision making process as I go.

```
# since this notebook is not in the main directory, use the here package
# to get proper file paths
library(here)
```

```
here() starts at /mnt/c/Users/jjk06/OneDrive/Desktop/2026_r_projs/hybrid_fitness_predict
```

```
library(ggplot2)
library(tidyr)
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

Creating a file of seed sizes

Seeds were planted 20 at a time into grids of 4 rows by 5 columns. Before planting the seeds, they were placed onto a grid as they were to be planted and a photo taken with a scale in the photo. The areas of the seeds were then measured from the photograph. Four researchers were working on the project, to get used to using imageJ and make sure there was no bias, the seeds were measured by all four researchers.

First, I'm reading in the four files and just calculate the mean seed area for each seed.

```
initials <- c("jk", "js", "lb", "rp")

# there were some notes etc. beyond the 4th column,
# so only take the first 4 columns
tmp <- read.csv(here(paste0("data/raw/seed_areas_",
                            initials[1], ".csv")))[,1:4]

tmp$researcher <- initials[1]

d <- tmp

for(i in 2:length(initials)){
  tmp <- read.csv(here(paste0("data/raw/seed_areas_",
                              initials[i], ".csv")))[,1:4]

  tmp$researcher <- initials[i]

  d <- rbind(d, tmp)
}


# calculate the mean of the four measurements
seed_size <- d %>%
  group_by(tray, block, ind) %>%
  summarise(seed_area = round(mean(area),3))
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by tray, block, and ind.
i Output is grouped by tray and block.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(tray, block, ind))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

Next, I am going to attach the cross id and parentage to each individual.

```
ids <- read.csv(here("data/raw/planting_ids.csv"))

# tray format is different in the two files, so need to change that to merge
ids <- ids %>%
  mutate(tray = gsub("LC", "lc_", tray)) %>%
  select(tray, block, cross, mother, father)

seed_size <- seed_size %>%
  left_join(ids) %>%
  mutate(cross = tolower(cross))
```

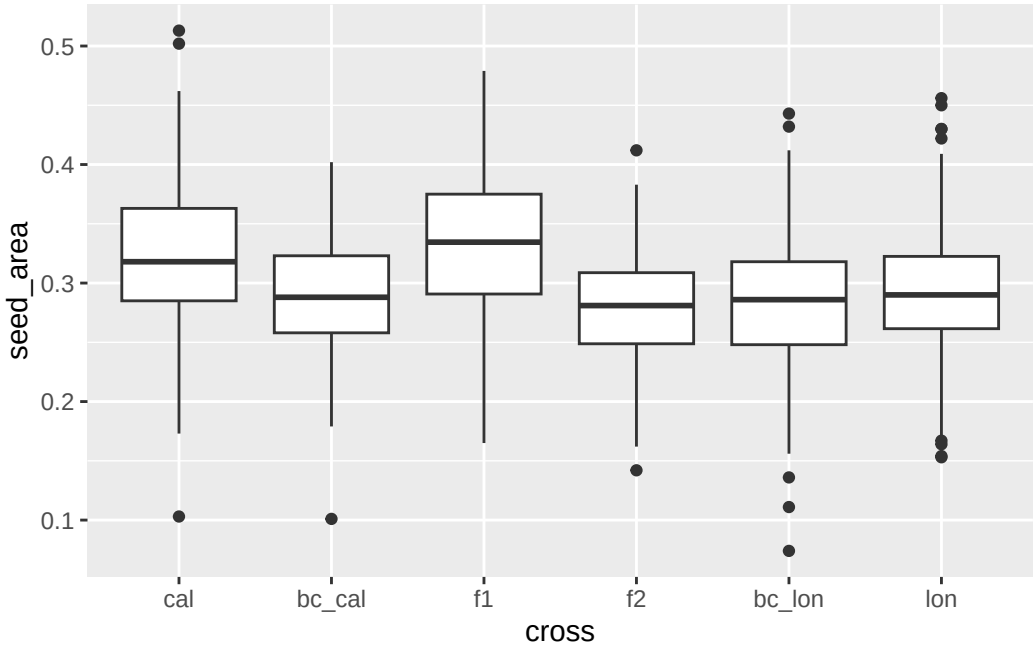
```
Joining with `by = join_by(tray, block)`
```

The above code takes the generates an object with seed sizes and cross information. It will be used in the script `scripts/wrangling.R` to generate the intermediate data file `reulsts/data/seed_size.csv`

Exploratory data analysis with seed size

First let's look at seed size vs cross type.

```
seed_size %>%
  mutate(cross = tolower(cross),
         cross = factor(cross,
                        levels = c("cal", "bc_cal", "f1",
                                   "f2", "bc_lon", "lon"))) %>%
  drop_na(seed_area) %>%
  ggplot(aes(x = cross, y = seed_area)) +
  geom_boxplot()
```



It looks like f1 and cal are the biggest with everything else being fairly similar. Let's just do a quick anova to see what that looks like.

```
summary(mod <- aov(seed_area ~ cross, data = seed_size))
```

```
              Df Sum Sq Mean Sq F value Pr(>F)
cross          5  0.594 0.11876   41.11 <2e-16 ***
Residuals   1370  3.957 0.00289
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
64 observations deleted due to missingness
```

```
TukeyHSD(mod)
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

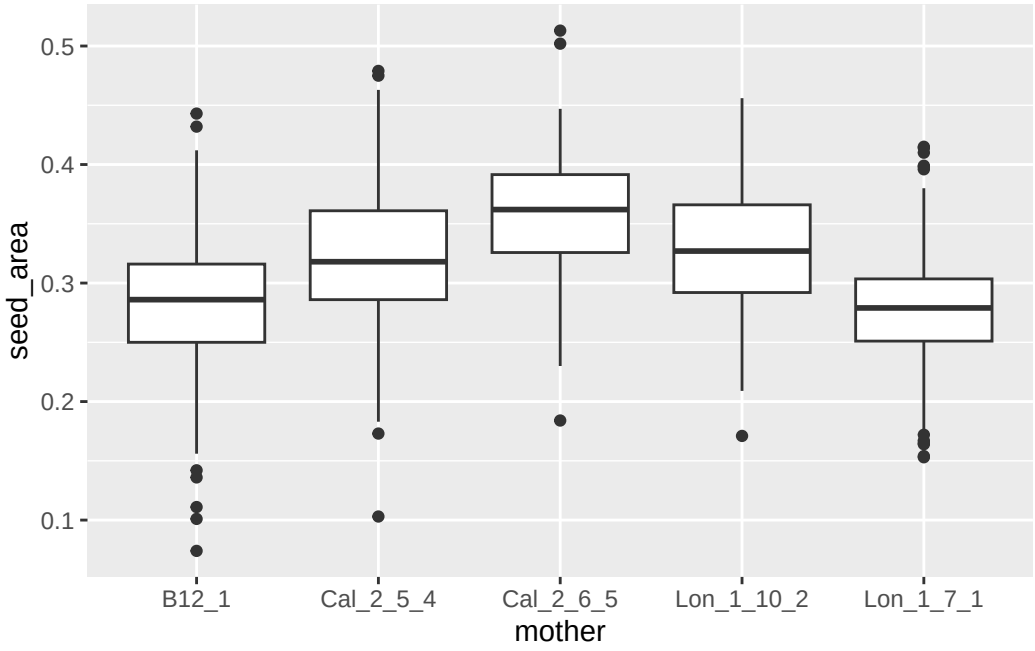
Fit: aov(formula = seed_area ~ cross, data = seed_size)

$cross
                       diff           lwr          upr     p adj
bc_lon-bc_cal -0.004700301 -0.0190972135  0.009696612 0.9383445
```

```
cal-bc_cal     0.036615225  0.0223096875  0.050920762 0.0000000
f1-bc_cal      0.044222433  0.0298717261  0.058573140 0.0000000
f2-bc_cal     -0.009266329 -0.0237426551  0.005209996 0.4487747
lon-bc_cal     0.004501934 -0.0098640582  0.018867925 0.9479315
cal-bc_lon     0.041315525  0.0270739589  0.055557092 0.0000000
f1-bc_lon      0.048922734  0.0346357954  0.063209672 0.0000000
f2-bc_lon     -0.004566029 -0.0189791416  0.009847085 0.9455309
lon-bc_lon     0.009202234 -0.0051000571  0.023504526 0.4427199
f1-cal         0.007607208 -0.0065876466  0.021802063 0.6452916
f2-cal        -0.045881554 -0.0602033949 -0.031559713 0.0000000
lon-cal       -0.032113291 -0.0463235986 -0.017902983 0.0000000
f2-f1         -0.053488762 -0.0678557217 -0.039121803 0.0000000
lon-f1        -0.039720499 -0.0539762784 -0.025464720 0.0000000
lon-f2         0.013768263 -0.0006139641  0.028150490 0.0697293
```

So it seems like there might be some real differences here. I'm not so sure that it is because of the hybrid embryo though. Seed size is at least partially, if not mostly a maternal trait, so lets look at seed size vs seed parent.

```
seed_size %>%
  mutate(cross = tolower(cross),
         cross = factor(cross,
                        levels = c("cal", "bc_cal", "f1",
                                   "f2", "bc_lon", "lon"))) %>%
  drop_na(seed_area) %>%
  ggplot(aes(x = mother, y = seed_area)) +
  geom_boxplot()
```



Looks like there are some differences, let's do a quick anova and see.

```
summary(mod <- aov(seed_area ~ mother, data = seed_size))
```

```
              Df Sum Sq Mean Sq F value Pr(>F)
mother         4  0.814 0.20339   74.61 <2e-16 ***
Residuals   1371  3.737 0.00273
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
64 observations deleted due to missingness
```

```
TukeyHSD(mod)
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

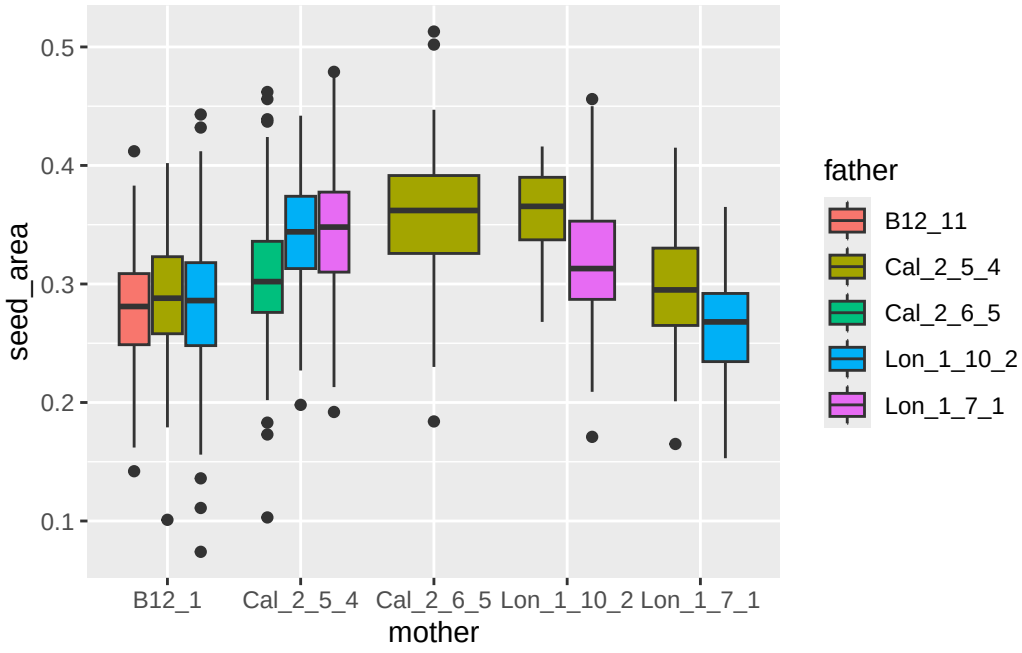
Fit: aov(formula = seed_area ~ mother, data = seed_size)

$mother
                            diff          lwr          upr     p adj
Cal_2_5_4-B12_1       0.039812279  0.029589976  0.050034582 0.0000000
Cal_2_6_5-B12_1       0.073819605  0.056768466  0.090870745 0.0000000
```

```
Lon_1_10_2-B12_1       0.046663473  0.034032604  0.059294341 0.0000000
Lon_1_7_1-B12_1       -0.004230057 -0.015936444  0.007476330 0.8612564
Cal_2_6_5-Cal_2_5_4    0.034007326  0.015697695  0.052316957 0.0000044
Lon_1_10_2-Cal_2_5_4   0.006851193 -0.007433066  0.021135453 0.6849996
Lon_1_7_1-Cal_2_5_4   -0.044042337 -0.057516042 -0.030568632 0.0000000
Lon_1_10_2-Cal_2_6_5  -0.027156133 -0.046911789 -0.007400476 0.0016930
Lon_1_7_1-Cal_2_6_5   -0.078049663 -0.097227426 -0.058871899 0.0000000
Lon_1_7_1-Lon_1_10_2  -0.050893530 -0.066274820 -0.035512240 0.0000000
```

It looks like there might be some real differences between the mothers. Most of the parentals were crossed with an opposite parent as well, so let's add in paternal information to see how the embryo genotype affects seed size.

```
seed_size %>%
  mutate(cross = tolower(cross),
         cross = factor(cross,
                        levels = c("cal", "bc_cal", "f1",
                                   "f2", "bc_lon", "lon"))) %>%
  drop_na(seed_area) %>%
  ggplot(aes(x = mother, y = seed_area, fill = father)) +
  geom_boxplot()
```



It seems that making crosses between parents increases seed size within the same mother, i.e., f1 embryos produce bigger seeds. The F1 mother produces the same size seeds whether crossed with another f1, lon, or, cal.

One last thing to do with seed size before moving on is that there is some missing data here. What happened is we tried to pre-apportion 20 seeds for each block ahead of time, but when we poured out the seeds or were arranging them to be photographed, we would sometimes loose a couple as they would jump around from static electricity or a misplaced sigh and then couldn't find them again. When that happened, we would leave the last places blank. So I just want to go through and figure out how many were missing from each block. Just placing it here so If I need it for quick reference later I can come find it.

```
seed_size %>%
  drop_na(seed_area) %>%
  group_by(tray, block) %>%
  summarise(n = n())
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by tray and block.
i Output is grouped by tray.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(tray, block))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
# A tibble: 72 x 3
# Groups:   tray [6]
   tray  block     n
   <chr> <int> <int>
 1 lc_1      1    15
 2 lc_1      2    20
 3 lc_1      3    17
 4 lc_1      4    20
 5 lc_1      5    20
 6 lc_1      6    20
 7 lc_1      7    20
 8 lc_1      8    20
 9 lc_1      9    20
10 lc_1     10    20
# i 62 more rows
```

Wrangling the germination timing data

I think the next set of data that needs to be wrangled is germination timing. The file `data/raw/germination_timing.csv` has observations of germination. The column names here are tray, block, ind, which correspond with the seed areas above. The rest of the columns are days that I checked for germination with the format 24263_16 (R adds an X that I'll have to remove). Where 24263 means the Julian date 263 for year 2024 and the 16 indicates the time the observation was taken. I think the time is probably too fine grained for this analysis, so I am going to remove it as well. For efficiency's sake, I entered a 1 if the plant had germinated and didn't enter anything if it hadn't, so no germination at a time point will be an NA

```
germ <- read.csv(here("data/raw/germination_timing.csv"))
head(germ)
```

```
  tray block ind X24263_16 X24264_17 X24265_14.5 X24266_14.5 X24267_13.5
1 lc_1     1   1        NA        NA          NA          NA          NA
2 lc_1     1   2        NA        NA          NA          NA          NA
3 lc_1     1   3        NA        NA          NA          NA          NA
4 lc_1     1   4        NA        NA          NA          NA          NA
5 lc_1     1   5        NA        NA          NA          NA          NA
6 lc_1     1   6        NA        NA          NA          NA          NA
  X24269_10 X24270_15.5 X24271_13.5 X24274_16.25 X24277_13.0
1         1           1           1            1           1
2        NA           1           1            1           1
3         1           1           1            1           1
4         1           1           1            1           1
5        NA          NA          NA           NA           1
6        NA          NA          NA           NA          NA
```

First thing is to make it a little more tidy.

```
(germ <- germ %>%
  pivot_longer(4:ncol(germ), names_to = "date", values_to = "germ") %>%
  # get rid of X in front of date, separate and get rid of the time
  # and make date numeric (also get rid of 24 since they are all
  # 2024)
  mutate(date = gsub("X24", "", date)) %>%
  separate(date, into = c("date", NA), sep = "_") %>%
  mutate(date = as.numeric(date)) %>%
  # now drop all the NAs when a germ wasn't observed and keep only the
  # earliest observed date, i.e., when it germinated
```

```
  drop_na(germ) %>%
  group_by(tray, block, ind) %>%
  filter(date == min(date)))
```

```
# A tibble: 904 x 5
# Groups:   tray, block, ind [904]
   tray  block   ind  date  germ
   <chr> <int> <int> <dbl> <int>
 1 lc_1      1     1   269     1
 2 lc_1      1     2   270     1
 3 lc_1      1     3   269     1
 4 lc_1      1     4   269     1
 5 lc_1      1     5   277     1
 6 lc_1      1     7   274     1
 7 lc_1      1     8   270     1
 8 lc_1      1     9   269     1
 9 lc_1      1    10   269     1
10 lc_1      1    11   274     1
# i 894 more rows
```

There is one more source of germination timing in the form of the seedling areas. We measured the seedling areas through the time period that germination was being taken, so if there is a measurement for the seedling it must have germinated. I'll blend these together to make sure I didn't miss anything.

```
areas <- read.csv(here("data/raw/seedling_areas.csv"))
head(areas)
```

```
  tray block ind area       time observer notes
1    1     1   1    1      24267_14.5       js
2    1     1   2    2      24267_14.5       js
3    1     1   3    3      24267_14.5       js
4    1     1   4    4      24267_14.5       js
5    1     1   5    5      24267_14.5       js
6    1     1   6    6      24267_14.5       js
```

The time column here is when the seedling area was measured. Again, I'm going to drop the hours and get rid of the 24. It looks like area was read in as a character, so I'm going to force it into numeric then drop the NAs

```r
(areas <- areas %>%
  # remove the 24 and hours
  separate(time, into = c("day", NA), sep = "_") %>%
  mutate(day = gsub("24","", day),
         day = as.numeric(day)) %>%
  # make area numeric and drop NAs
  mutate(area = as.numeric(area)) %>%
  drop_na(area) %>%
  # now get the earliest day for each individual
  group_by(tray, block, ind) %>%
  filter(day == min(day)) %>%
  # since multiple researchers measured each day/seedling combo, there
  # are multiple instances of each day, so filter down to just one
  ungroup() %>%
  filter(!duplicated(paste0(tray, "_", block, "_", ind))))
```

```
Warning: There was 1 warning in `mutate()`.
i In argument: `area = as.numeric(area)`.
Caused by warning:
! NAs introduced by coercion
```

```
# A tibble: 920 x 7
    tray block   ind  area   day observer notes
   <int> <int> <int> <dbl> <dbl> <chr>    <chr>
 1     1     1    14 1.27    267 js       ""
 2     1     2     3 2.31    267 js       ""
 3     1     2    12 0.558   267 js       ""
 4     1     3     4 3.16    267 js       ""
 5     1     3     5 1.49    267 js       ""
 6     1     3     8 3.31    267 js       ""
 7     1     3    14 1.09    267 js       ""
 8     1     4     3 4.26    267 js       ""
 9     1     4     4 2.58    267 js       ""
10     1     4     6 2.54    267 js       ""
# i 910 more rows
```

Now I'm going to join the two data sets and select the earliest observation.

```r
(germ_day <- areas %>%
  mutate(tray = paste0("lc_", tray)) %>%
  left_join(germ, by = join_by(tray, block, ind)) %>%
```

```
  pivot_longer(c(day, date), values_to = "germ_day") %>%
  drop_na(germ_day) %>%
  group_by(tray, block, ind) %>%
  filter(germ_day == min(germ_day)) %>%
  filter(!duplicated(tray, block, ind)) %>%
  select(tray, block, ind, germ_day))
```

```
# A tibble: 920 x 4
# Groups:   tray, block, ind [920]
   tray  block   ind germ_day
   <chr> <int> <int>    <dbl>
 1 lc_1      1    14      265
 2 lc_1      2     3      265
 3 lc_1      2    12      266
 4 lc_1      3     4      263
 5 lc_1      3     5      266
 6 lc_1      3     8      264
 7 lc_1      3    14      265
 8 lc_1      4     3      263
 9 lc_1      4     4      265
10 lc_1      4     6      265
# i 910 more rows
```

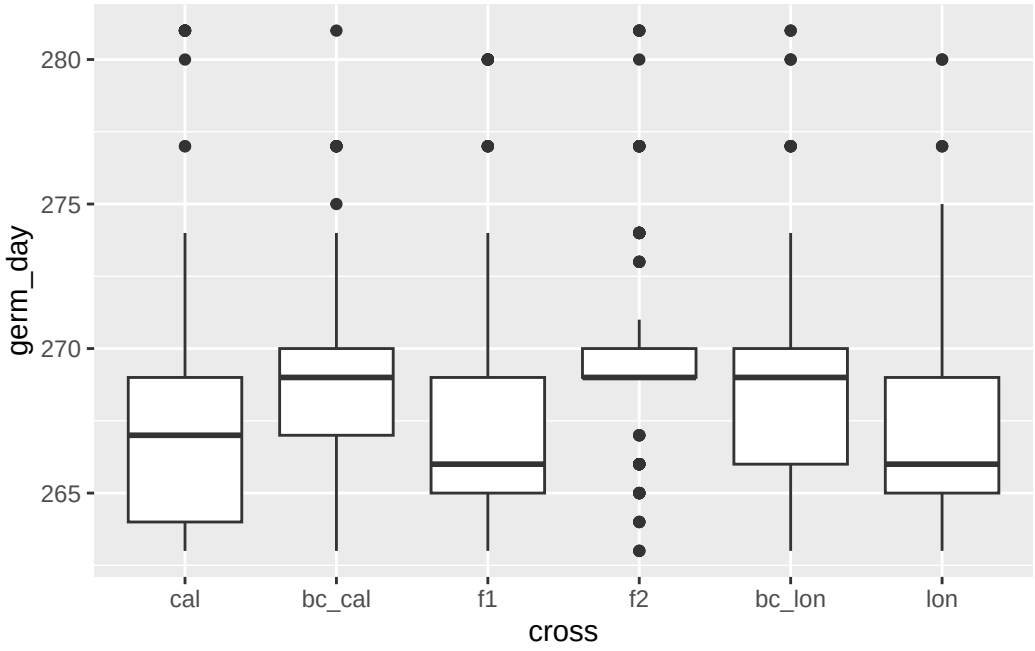
Exploratory analysis of germination timing

Ok, now that we have some germ times, let's join it with the planting ids to get cross and parentage info and do some exploratory analysis.

First, let's look at germination time vs cross.

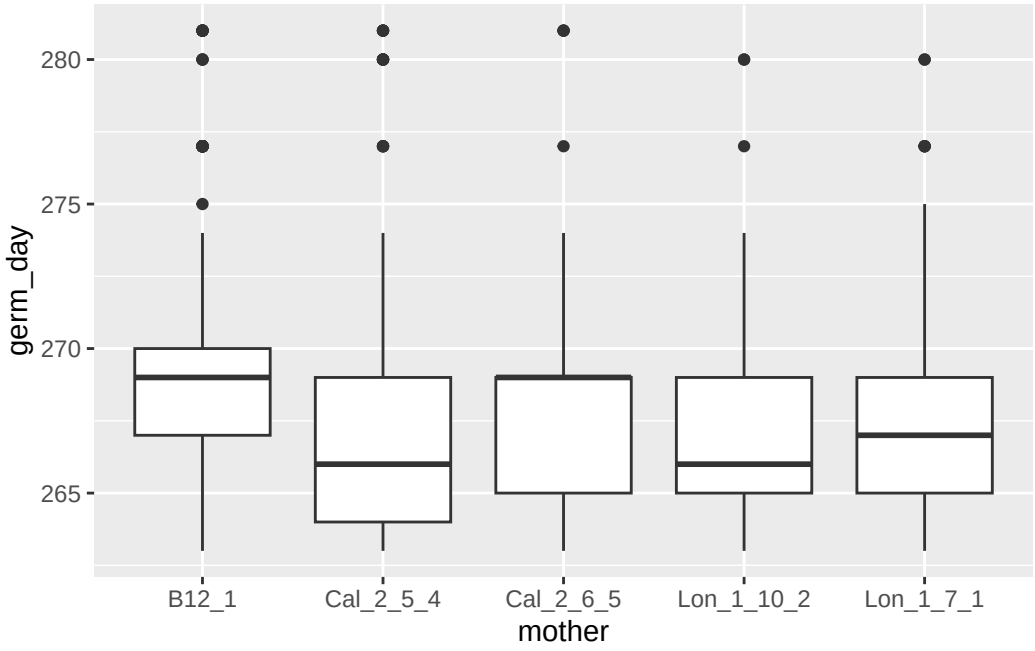
```
germ_day <- germ_day %>%
  left_join(ids, by = join_by(tray, block)) %>%
  mutate(cross = tolower(cross),
         cross = factor(cross,
                        levels = c("cal", "bc_cal", "f1",
                                   "f2", "bc_lon", "lon")))

germ_day %>%
  ggplot(aes(x = cross, y = germ_day)) +
  geom_boxplot()
```



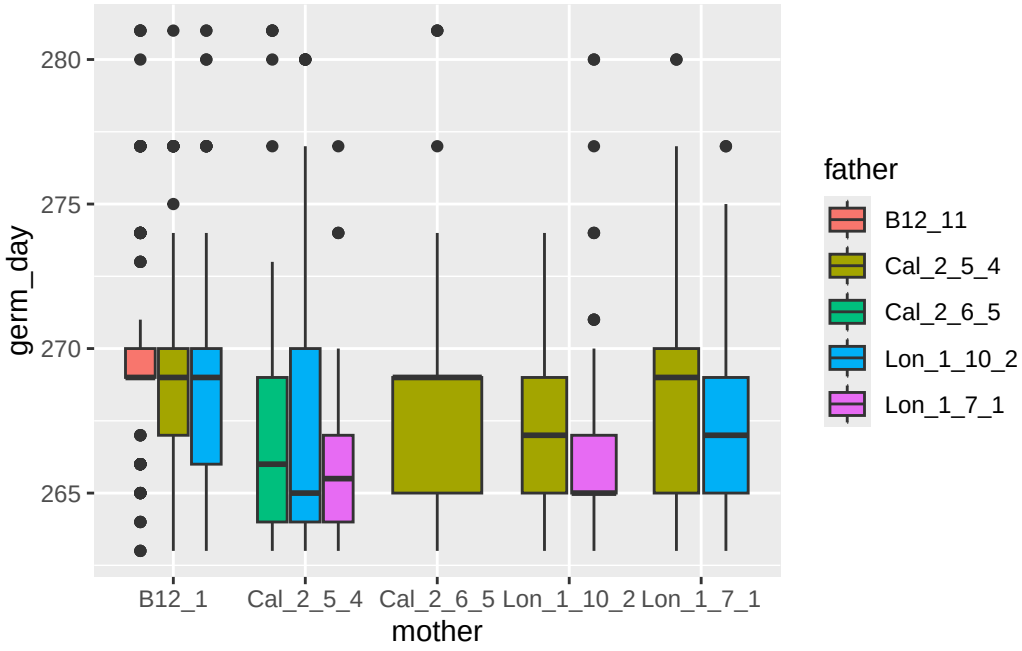
It looks like the back crosses and f2 take longer to germinate than the f1s and parentals. Let's look at it by seed parent.

```
germ_day %>%
  ggplot(aes(x = mother, y = germ_day)) +
  geom_boxplot()
```



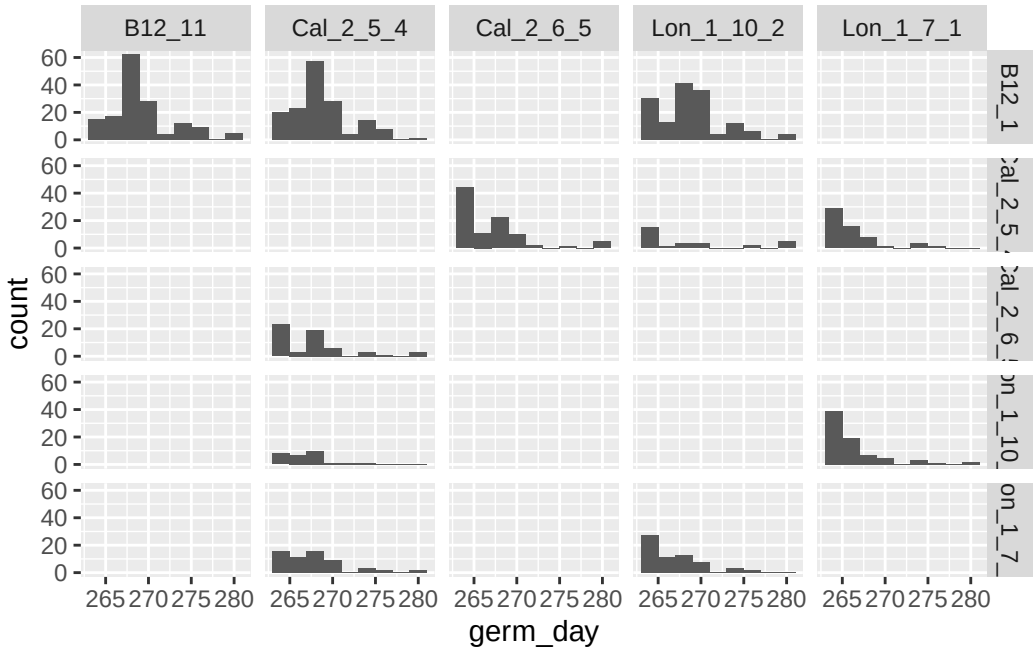
Not sure what to think abut this, but let's look at it breaking out the individual crosses (i.e. include the father information)

```
germ_day %>%
  ggplot(aes(x = mother, y = germ_day, fill = father)) +
  geom_boxplot()
```



The f2 pattern looks really weird so let's look at the distribution of germ days.

```
germ_day %>%
  ggplot(aes(x = germ_day)) +
  geom_histogram(binwidth = 2) +
  facet_grid(mother ~ father)
```



Is there a relationship between seed size and germination timing?

```
seed_size <- seed_size %>% ungroup()
germ_day <- germ_day %>% ungroup()

d <- seed_size %>%
  left_join(germ_day)
```

```
Joining with `by = join_by(tray, block, ind, cross, mother, father)`
```

```
# since the experiment started on Julian day 260, subtract that from germ
# day to get the actual germ_day.

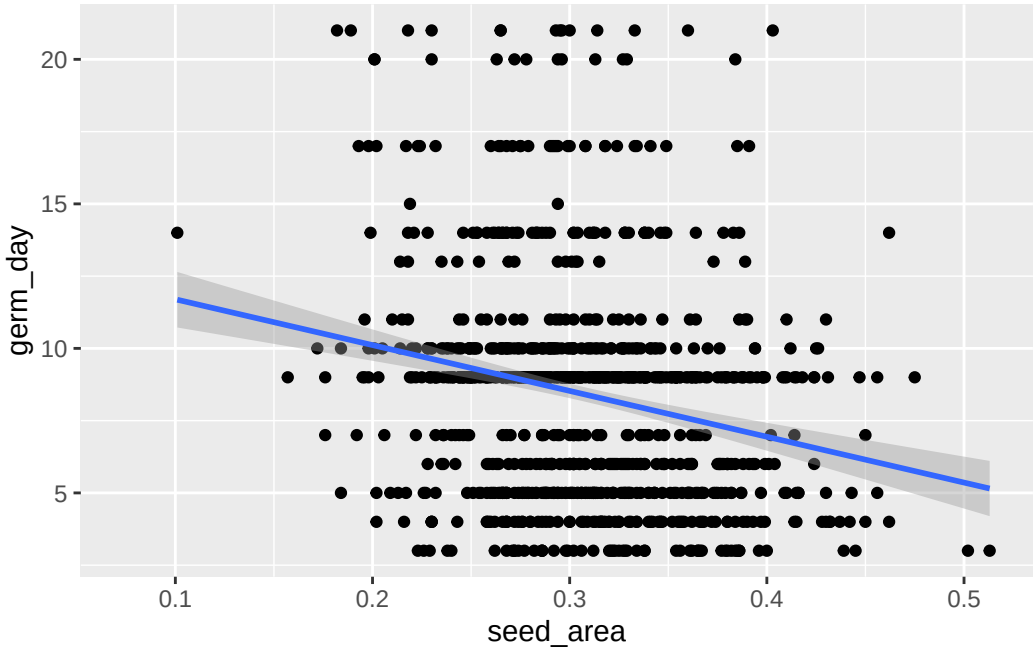
d <- d %>%
  mutate(germ_day = germ_day - 260)

d %>%
  ggplot(aes(x = seed_area, y = germ_day)) +
  geom_point() +
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 522 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 522 rows containing missing values or values outside the scale range
(`geom_point()`).
```



There definitely seems to be a negative relationship between seed size and germination timing.

I'm going to add some of the above code to the `wrangling.R` script so that the output includes germination timing as well.

Probability of Germination

Next I want to look at the probability of germination and check out the relationship between size and cross type.

```
(d <- d %>%
  # first drop the NA in seed_area because they were places where
  # there are no seeds.
  drop_na(seed_area) %>%
  # convert germ day into whether or not a plant germinated. NAs are
  # non-germinants
```

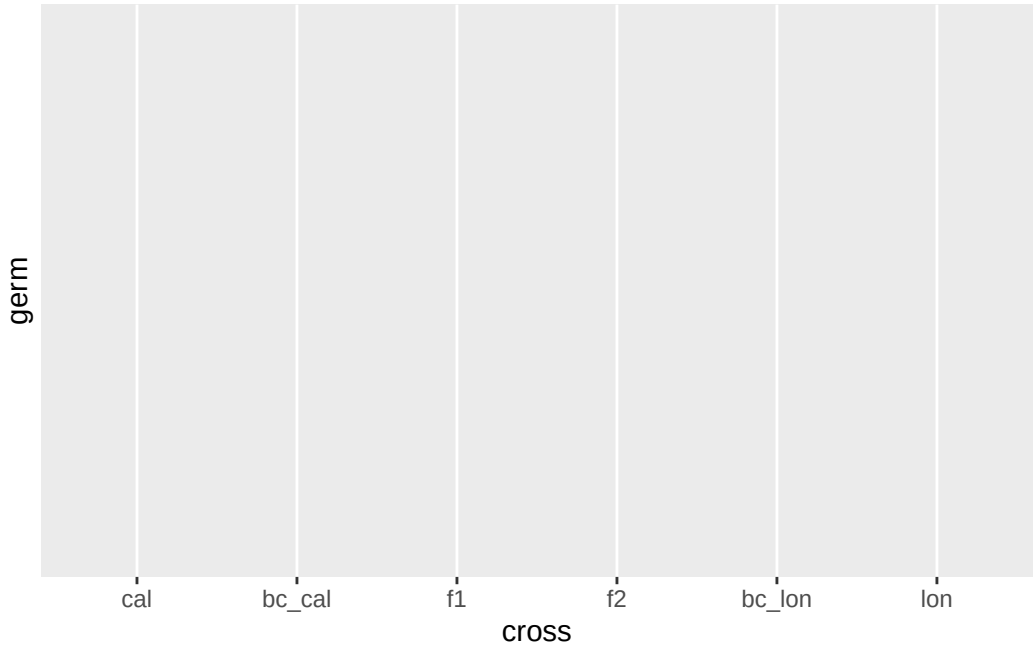
```
  mutate(germ = ifelse(is.na(germ_day), 0, 1)) %>%
  mutate(cross = factor(cross, levels = c("cal", "bc_cal", "f1", "f2",
                                          "bc_lon", "lon"))))
```

```
# A tibble: 1,376 x 9
   tray  block   ind seed_area cross mother father germ_day  germ
   <chr> <int> <int>     <dbl> <fct> <chr>  <chr>     <dbl> <dbl>
 1 lc_1      1     1     0.322 f2    B12_1  B12_11        9     1
 2 lc_1      1     2     0.287 f2    B12_1  B12_11       10     1
 3 lc_1      1     3     0.286 f2    B12_1  B12_11        9     1
 4 lc_1      1     4     0.274 f2    B12_1  B12_11        9     1
 5 lc_1      1     5     0.318 f2    B12_1  B12_11       17     1
 6 lc_1      1     6     0.239 f2    B12_1  B12_11       NA     0
 7 lc_1      1     7     0.253 f2    B12_1  B12_11       14     1
 8 lc_1      1     8     0.228 f2    B12_1  B12_11       10     1
 9 lc_1      1     9     0.33  f2    B12_1  B12_11        9     1
10 lc_1      1    10     0.313 f2    B12_1  B12_11        9     1
# i 1,366 more rows
```

Let's look at germination by cross first.

```
d %>%
  ggplot(aes(x = cross, y = germ)) +
  # use bootstrap to plot the means and confidence intervals.
  stat_summary(fun.data = mean_cl_boot, geom = "point") +
  stat_summary(fun.data = mean_cl_boot, geom = "errorbar")
```

```
Warning: Computation failed in `stat_summary()`.
Computation failed in `stat_summary()`.
Caused by error in `fun.data()`:
! The package "Hmisc" is required.
```

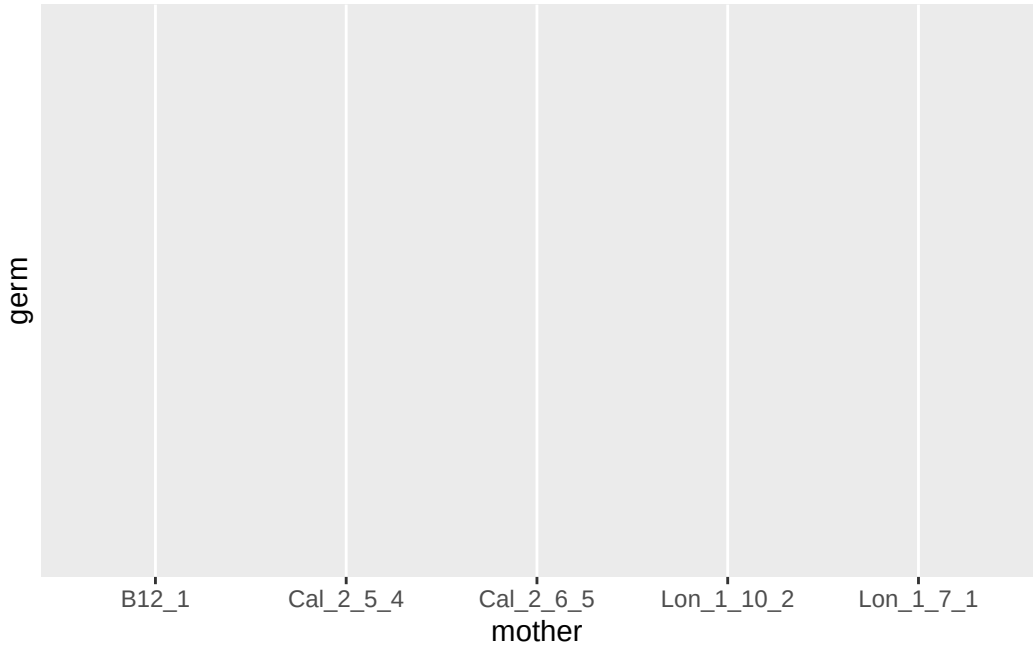


It looks like the f1s have a higher germination percentage than the rest, but not so different between everything else maybe.

Let's look at by mother now.

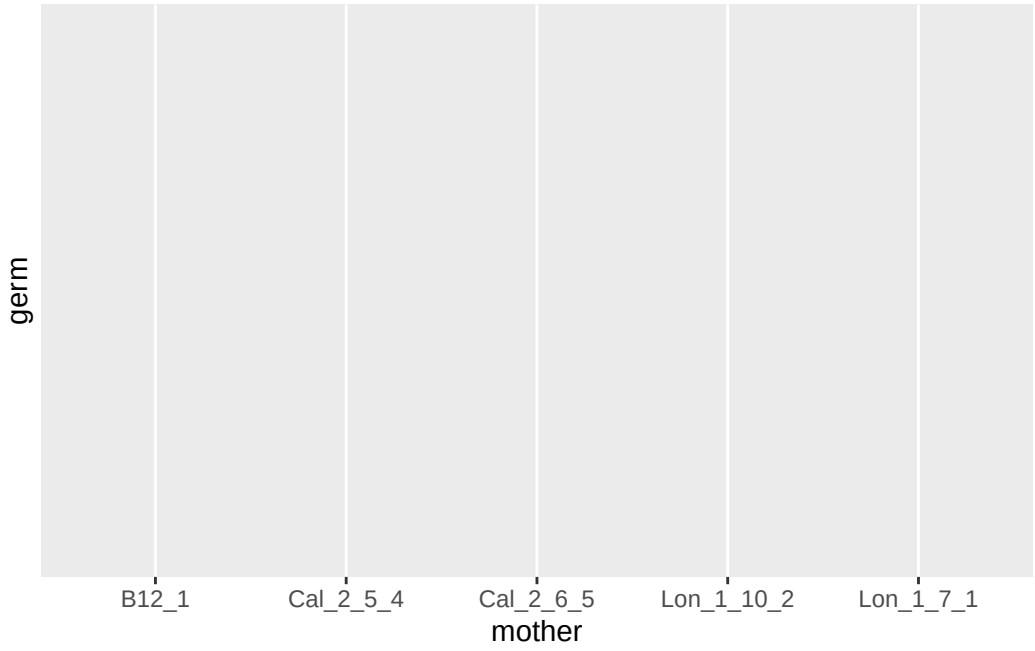
```
d %>%
  ggplot(aes(x = mother, y = germ)) +
  # use bootstrap to plot the means and confidence intervals.
  stat_summary(fun.data = mean_cl_boot, geom = "point") +
  stat_summary(fun.data = mean_cl_boot, geom = "errorbar")
```

```
Warning: Computation failed in `stat_summary()`.
Computation failed in `stat_summary()`.
Caused by error in `fun.data()`:
! The package "Hmisc" is required.
```



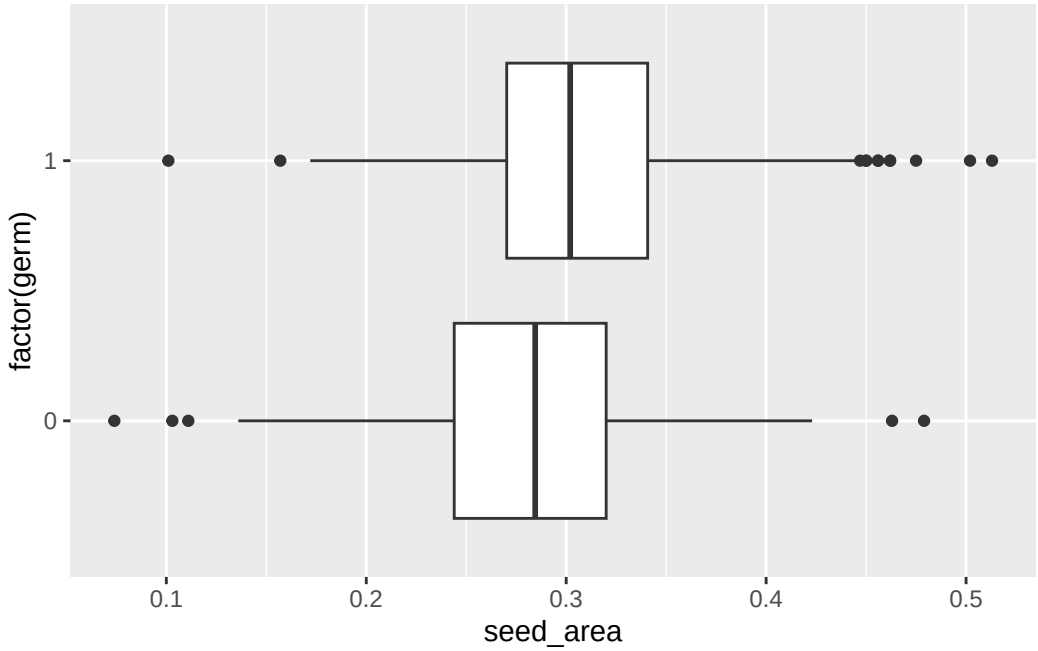

```
pos <- position_jitterdodge(jitter.width = .2, jitter.height = 0)
d %>%
  ggplot(aes(x = mother, y = germ, color = father)) +
  # use bootstrap to plot the means and confidence intervals.
  stat_summary(fun.data = mean_cl_boot, geom = "point",
               position = pos) +
  stat_summary(fun.data = mean_cl_boot, geom = "errorbar",
               position = pos)
```

```
Warning: Computation failed in `stat_summary()`.
Computation failed in `stat_summary()`.
Caused by error in `fun.data()`:
! The package "Hmisc" is required.
```



This is ugly, but I don't want to spend more time on it now. The pattern looks very much like the same plot made with seed size, so I'm expecting seed size should be predictive of germination. Let's look at that next.

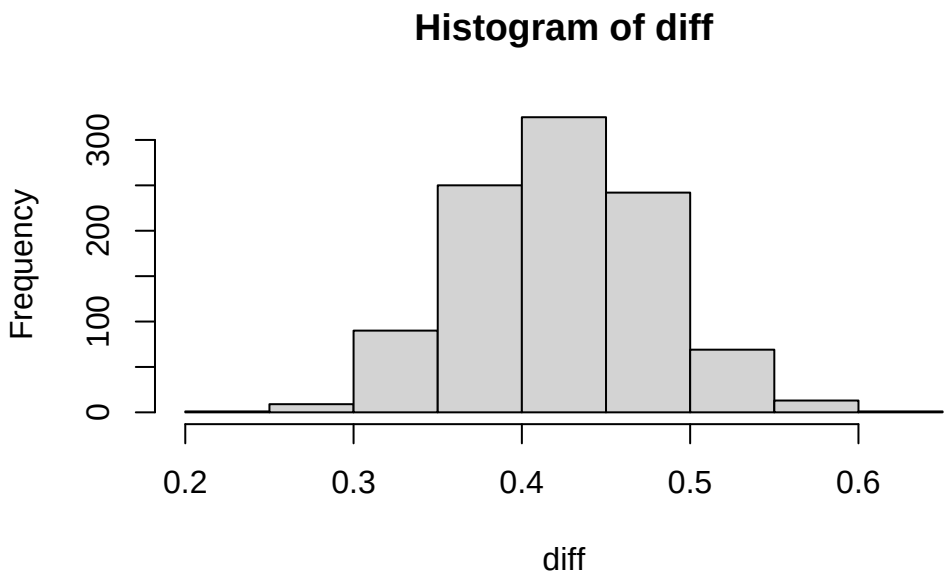
```
d %>%
  ggplot(aes(x = seed_area, y = factor(germ))) +
  geom_boxplot()
```



Sure looks like the germinants were bigger. Let's do just a quick bootstrap to test the difference in the means.

```
germ <- d$germ
# standardize seed area for ease of interpretation
seed_area <- (d$seed_area - mean(d$seed_area))/sd(d$seed_area)
diff <- c()
for(i in 1:1000){
  samps <- sample(1:nrow(d), replace = T)
  g <- germ[samps]
  s <- seed_area[samps]
  diff[i] <- mean(s[g==1]) - mean(s[g==0])
}

hist(diff)
```



So the germinating seeds are roughly 0.4 standard deviations larger than the non-germinants–on average.

Seedling Sizes

The next set of data that needs to be wrangled is the seedling area measurements. At each timepoint, the seedlings were photographed and measured by two people (or will be, I still need to finish my measurements…). It might not be the absolute best approach—probably should model the differences—but I'm going to average across the multiple observations of each seedling at each timepoint.

```
seedlings <- read.csv(here("data/raw/seedling_areas.csv"))

(seedlings <- seedlings %>%
  # make area numeric, then drop the NAs to get just observations.
  mutate(area = as.numeric(area)) %>%
  drop_na(area) %>%
  # the time stamp is the same as above for germination, just going to
  # make it the day
  separate(time, into = c("day", NA), sep = "_") %>%
  mutate(day = gsub("24", "", day),
         day = as.numeric(day)) %>%
```

```r
  # now average each seedling time.
  group_by(tray, block, ind, day) %>%
  summarise(area = round(mean(area),3)) %>%
  ungroup() %>%
  mutate(tray = paste0("lc_", tray)) %>%
  # join with the germ day data to get cross information as well
  # as germination time to calculate age.
  left_join(germ_day, by = join_by(tray, block, ind)) %>%
  mutate(age = day - germ_day) %>%
  select(tray, block, ind, area, age, cross, mother, father))
```

```
Warning: There was 1 warning in `mutate()`.
i In argument: `area = as.numeric(area)`.
Caused by warning:
! NAs introduced by coercion
```

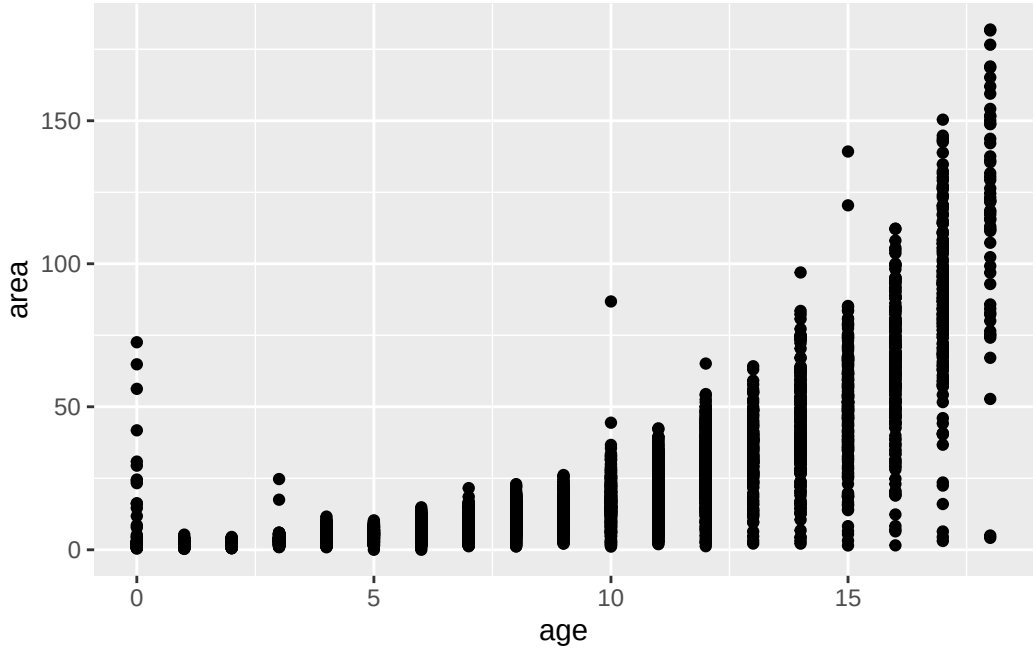
```
`summarise()` has regrouped the output.
i Summaries were computed grouped by tray, block, ind, and day.
i Output is grouped by tray, block, and ind.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(tray, block, ind, day))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
# A tibble: 5,113 x 8
   tray  block   ind   area   age cross mother father
   <chr> <int> <int>  <dbl> <dbl> <fct> <chr>  <chr>
 1 lc_1      1     1  0.989     1 f2    B12_1  B12_11
 2 lc_1      1     1  4.50      4 f2    B12_1  B12_11
 3 lc_1      1     1  6.81      6 f2    B12_1  B12_11
 4 lc_1      1     1  8.93      8 f2    B12_1  B12_11
 5 lc_1      1     1 14.8      11 f2    B12_1  B12_11
 6 lc_1      1     1 18.3      12 f2    B12_1  B12_11
 7 lc_1      1     2  5.92      3 f2    B12_1  B12_11
 8 lc_1      1     2  8.42      5 f2    B12_1  B12_11
 9 lc_1      1     2 12.6       7 f2    B12_1  B12_11
10 lc_1      1     2 24.0      10 f2    B12_1  B12_11
# i 5,103 more rows
```

Add the above code to `wrangling.R` to create the seedling area file.

Some exploratory analysis of growth.

```
seedlings %>%
  ggplot(aes(x = age, y = area)) +
  geom_point()
```



There are some very big seedlings at timepoint 0. Those are almost certainly errors. I'm eventually going to remove all the time 0 anyway because the plants are still unfolding on the germination day, so it is misleading.Let's filter for all the age 0's and see what's going on.

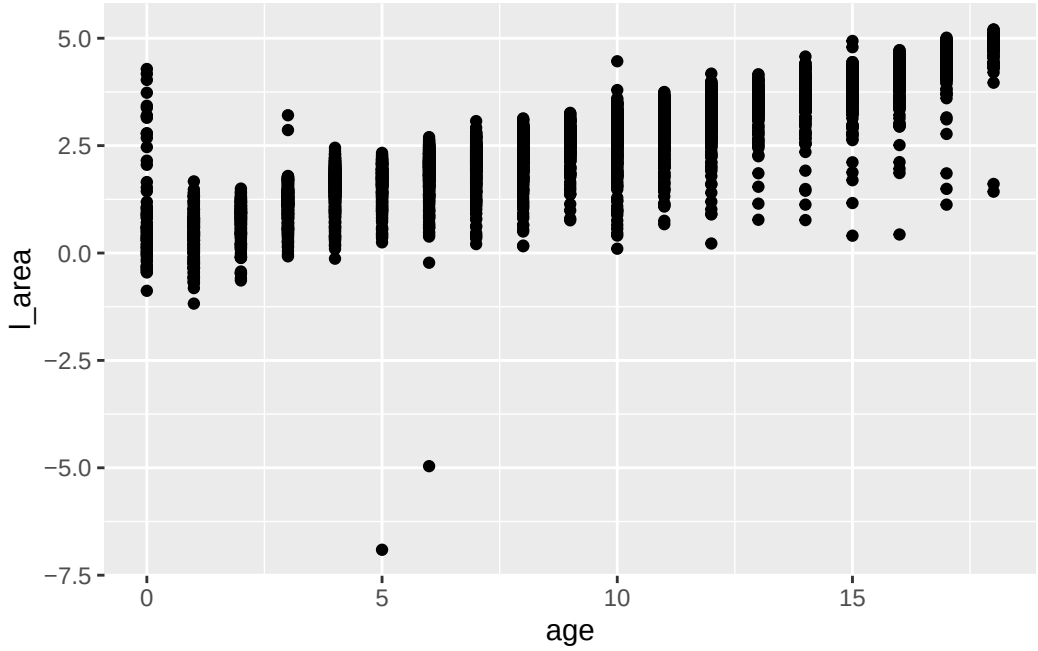
```
seedlings %>%
  filter(age == 0)
```

```
# A tibble: 70 x 8
   tray  block   ind  area   age cross  mother     father
   <chr> <int> <int> <dbl> <dbl> <fct>  <chr>      <chr>
 1 lc_1      1     5 2.49      0 f2     B12_1      B12_11
 2 lc_1      5     5 2.81      0 f1     Cal_2_5_4  Lon_1_7_1
 3 lc_1      6     2 0.678     0 cal    Cal_2_6_5  Cal_2_5_4
 4 lc_1      6    16 1.03      0 cal    Cal_2_6_5  Cal_2_5_4
 5 lc_1      7     2 4.22      0 f2     B12_1      B12_11
```

```
 6 lc_1      9     5 1.13      0 lon    Lon_1_7_1  Lon_1_10_2
 7 lc_1     10     1 1.61      0 bc_cal B12_1      Cal_2_5_4
 8 lc_1     11     9 1.22      0 f1     Lon_1_10_2 Cal_2_5_4
 9 lc_1     12    12 0.979     0 bc_lon B12_1      Lon_1_10_2
10 lc_2      2     6 0.415     0 bc_lon B12_1      Lon_1_10_2
# i 60 more rows
```

Let's plot log of area and see what that looks like.

```
seedlings %>%
  mutate(l_area = log(area)) %>%
  ggplot(aes(x = age, y = l_area)) +
  geom_point()
```



Still not sure what is going on with those big points at timepoint 0, I think maybe a date was marked incorrectly for them so they are not really timepoint 0. Whatever it is, those individuals don't seem to be outliers for the rest of the time, so I think it will be fine when I remove the timepoint 0 measurements.

There are also a couple positive outliers on day 3 and 10, and some big negative outliers on days 5 and 6. I'm going to check them out, they are probabaly mismeasurments and I will just remove those particular observations.

```
seedlings %>%
  mutate(l_area = log(area)) %>%
  filter(age == 3) %>%
  filter(l_area > 2.5)
```

```
# A tibble: 2 x 9
  tray  block   ind  area   age cross mother    father     l_area
  <chr> <int> <int> <dbl> <dbl> <fct> <chr>     <chr>       <dbl>
1 lc_2      3    13  17.5     3 f1    Lon_1_7_1 Cal_2_5_4    2.86
2 lc_6      9    11  24.7     3 f1    Cal_2_5_4 Lon_1_10_2   3.21
```

```
# so at age 3, it is lc2-3-13 and lc6-9-11, let's plot those two

seedlings %>%
  mutate(l_area = log(area)) %>%
  filter(tray == "lc_2" & block == 3 & ind == 13 |
           tray == "lc_6" & block == 9 & ind == 11)
```

```
# A tibble: 4 x 9
  tray  block   ind  area   age cross mother    father     l_area
  <chr> <int> <int> <dbl> <dbl> <fct> <chr>     <chr>       <dbl>
1 lc_2      3    13 17.5      3 f1    Lon_1_7_1 Cal_2_5_4   2.86
2 lc_2      3    13  2.78     4 f1    Lon_1_7_1 Cal_2_5_4   1.02
3 lc_6      9    11 24.7      3 f1    Cal_2_5_4 Lon_1_10_2  3.21
4 lc_6      9    11  2.60     4 f1    Cal_2_5_4 Lon_1_10_2  0.954
```

Ok, so they are smaller on day 4 than day 3, which negative growth is possible, but since they are outliers on day 3, I think this is some sort of recording error and I'm going to remove these individuals.

Now to look at thos individuals that are small at ages 5 and 6 and see if there is something systematic or if they are just wierd measurements.

```
seedlings %>%
  mutate(l_area = log(area)) %>%
  filter(l_area < -2.5)
```

```
# A tibble: 2 x 9
  tray  block   ind  area   age cross mother    father    l_area
  <chr> <int> <int> <dbl> <dbl> <fct> <chr>     <chr>      <dbl>
1 lc_3      3     4 0.001     5 f1    Lon_1_7_1 Cal_2_5_4  -6.91
2 lc_3     12    16 0.007     6 cal   Cal_2_5_4 Cal_2_6_5  -4.96
```

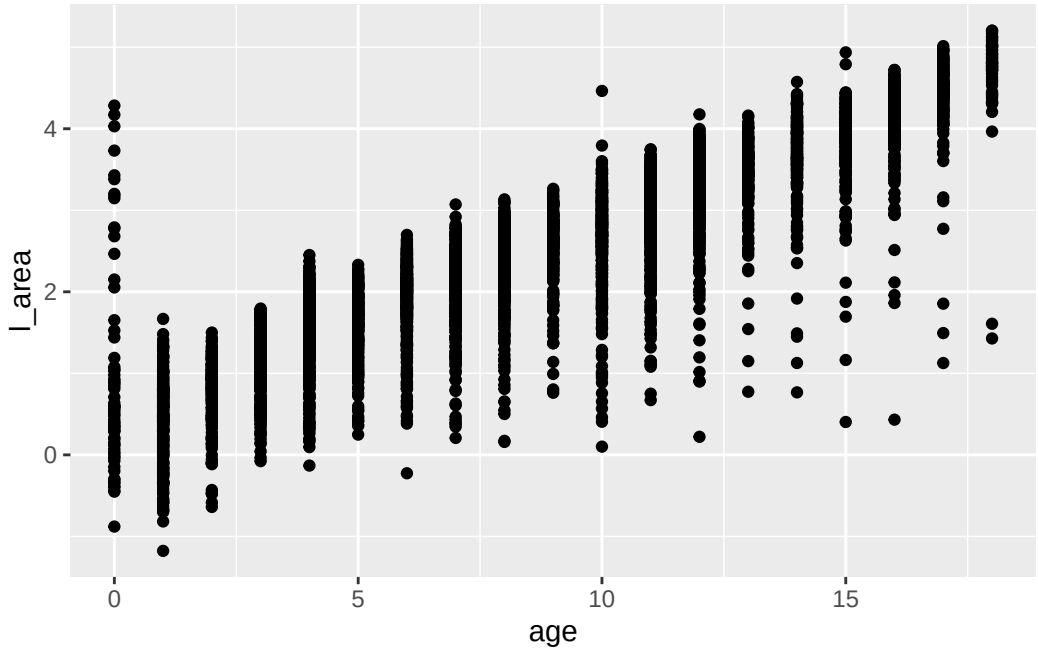
```
# it is lc_3-3-4 and lc3-12-16
seedlings %>%
  mutate(l_area = log(area)) %>%
  filter(tray == "lc_3" & block == 3 & ind == 4 |
           tray == "lc_3" & block == 12 & ind == 16)
```

```
# A tibble: 14 x 9
   tray  block   ind   area   age cross mother    father    l_area
   <chr> <int> <int>  <dbl> <dbl> <fct> <chr>     <chr>      <dbl>
 1 lc_3      3     4  1.79      2 f1    Lon_1_7_1 Cal_2_5_4  0.582
 2 lc_3      3     4  0.001     5 f1    Lon_1_7_1 Cal_2_5_4 -6.91
 3 lc_3      3     4 10.6       8 f1    Lon_1_7_1 Cal_2_5_4  2.36
 4 lc_3      3     4 15.4      10 f1    Lon_1_7_1 Cal_2_5_4  2.74
 5 lc_3      3     4 23.6      12 f1    Lon_1_7_1 Cal_2_5_4  3.16
 6 lc_3      3     4 46.5      15 f1    Lon_1_7_1 Cal_2_5_4  3.84
 7 lc_3      3     4 57.7      16 f1    Lon_1_7_1 Cal_2_5_4  4.06
 8 lc_3     12    16  2.90      3 cal   Cal_2_5_4 Cal_2_6_5  1.06
 9 lc_3     12    16  0.007     6 cal   Cal_2_5_4 Cal_2_6_5 -4.96
10 lc_3     12    16 14.2       9 cal   Cal_2_5_4 Cal_2_6_5  2.65
11 lc_3     12    16 20.1      11 cal   Cal_2_5_4 Cal_2_6_5  3.00
12 lc_3     12    16 30.8      13 cal   Cal_2_5_4 Cal_2_6_5  3.43
13 lc_3     12    16 54.7      16 cal   Cal_2_5_4 Cal_2_6_5  4.00
14 lc_3     12    16 70.4      17 cal   Cal_2_5_4 Cal_2_6_5  4.25
```

Ok, these are clearly just bad measurements for some reason. I will remove the single observations.

Make those filters and then re-check.

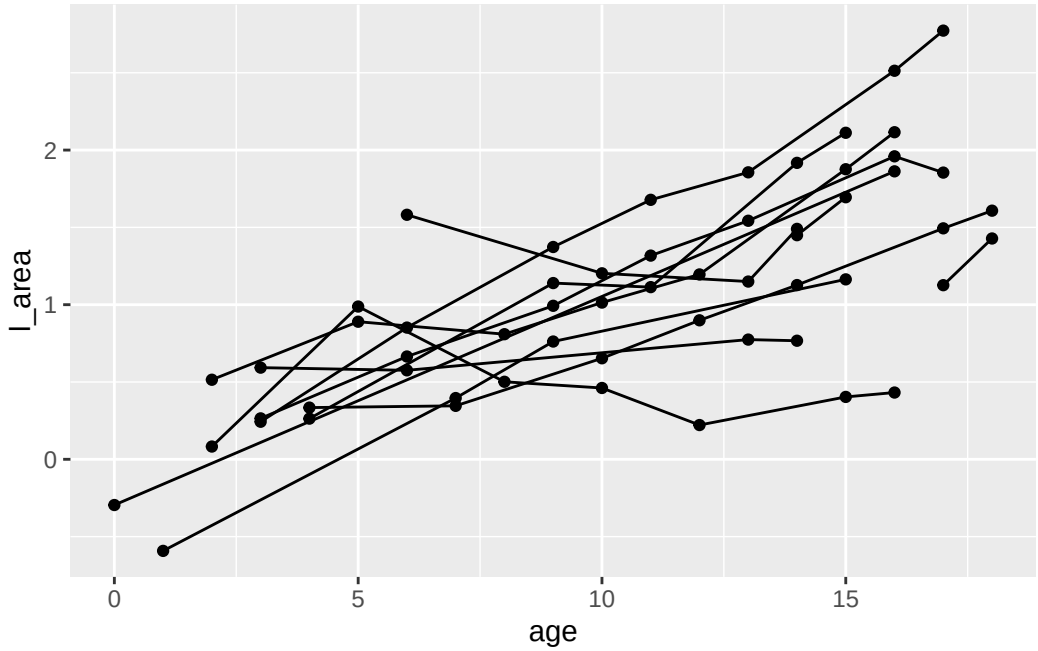
```
seedlings %>%
  mutate(l_area = log(area)) %>%
  filter(paste0(tray, block, ind) != "lc_2313") %>%
  filter(paste0(tray, block, ind) != "lc_6911") %>%
  filter(l_area > -2.5) %>%
  ggplot(aes(x = age, y = l_area)) +
  geom_point()
```



Let's take a peek at those small individuals at the bigger sizes to see what's going on there.

```
id <- seedlings %>%
  mutate(l_area = log(area)) %>%
  filter(age > 12 & l_area < 2) %>%
  mutate(id = paste0(tray, block, ind)) %>%
  pull(id)

seedlings %>%
  mutate(l_area = log(area)) %>%
  filter(paste0(tray, block, ind) %in% id) %>%
  ggplot(aes(x = age, y = l_area, group = paste0(tray, block, ind))) +
  geom_point() +
  geom_line()
```



For the most part, those look fine, but there are a couple that for some reason only have one or two observations, but given that they are later ages, that shouldn't happen. Checking into it.

```
seedlings %>%
  mutate(l_area = log(area)) %>%
  filter(paste0(tray, block, ind) %in% id) %>%
  group_by(tray, block, ind) %>%
  summarise(n = n(),
            age_max = max(age)) %>%
  filter(n <= 2)
```

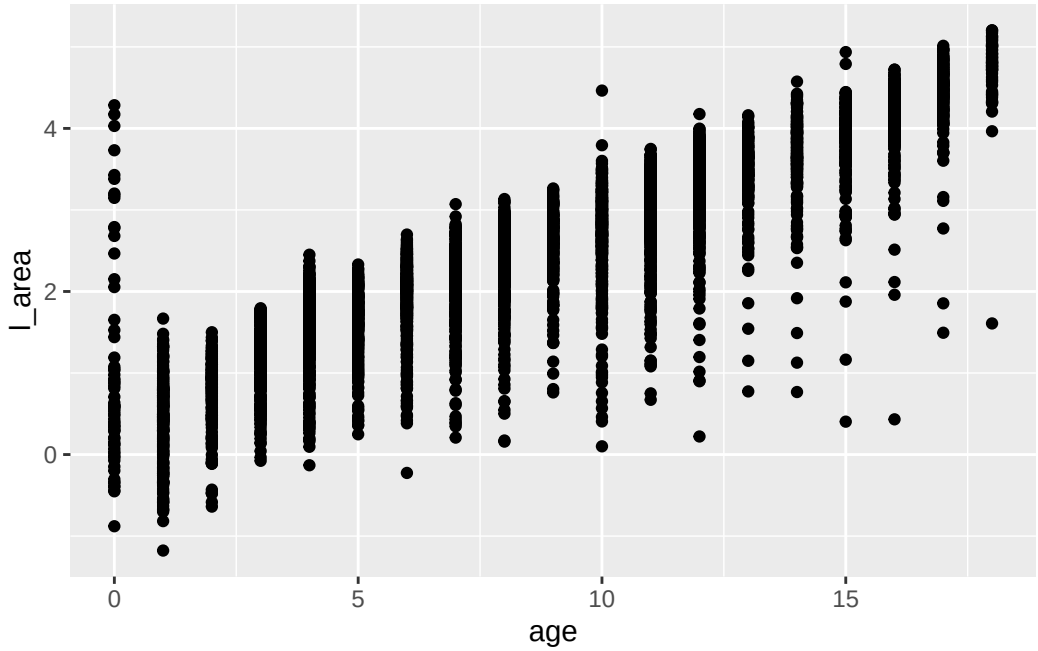
```
`summarise()` has regrouped the output.
i Summaries were computed grouped by tray, block, and ind.
i Output is grouped by tray and block.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(tray, block, ind))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.

# A tibble: 4 x 5
# Groups:   tray, block [4]
  tray  block   ind     n age_max
  <chr> <int> <int> <int>   <dbl>
1 lc_2      1    12     2      15
```

```
2 lc_4      9    12     2      18
3 lc_5      1    17     1      16
4 lc_5     11     7     1       0
```

Ok, these should have more observations given the age, so I'm going to add them to the prune list.

```
seedlings %>%
  mutate(l_area = log(area)) %>%
  filter(!(paste0(tray, block, ind) %in% c("lc_2313", "lc_6911",
                                           "lc_2112", "lc_4912",
                                           "lc_5117", "lc_5117"))) %>%
  filter(l_area > -2.5) %>%
  ggplot(aes(x = age, y = l_area)) +
  geom_point()
```



I'm just going to check out those big ones on day zero again, just to see if I can figure out what's going on and save them.

```
big_at_0 <- seedlings %>%
  mutate(l_area = log(area)) %>%
  filter(age < 1 & l_area > 2) %>%
  mutate(id = paste0(tray, "_", block, "_", ind)) %>%
```

```
  pull(id)

seedlings %>%
  mutate(l_area = log(area)) %>%
  filter(paste0(tray, "_", block, "_", ind) %in% big_at_0)
```

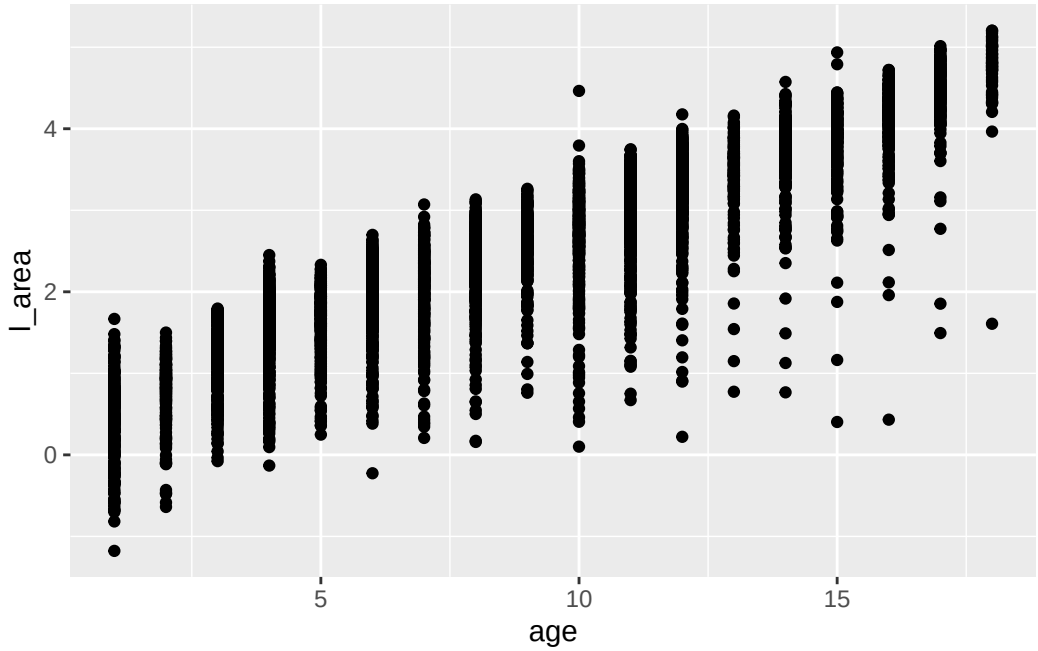
```
# A tibble: 15 x 9
   tray  block   ind  area   age cross mother     father     l_area
   <chr> <int> <int> <dbl> <dbl> <fct> <chr>      <chr>       <dbl>
 1 lc_2      7    17 72.6      0 lon   Lon_1_10_2 Lon_1_7_1    4.28
 2 lc_3      3     2 14.6      0 f1    Lon_1_7_1  Cal_2_5_4    2.68
 3 lc_4      5     9 56.3      0 cal   Cal_2_5_4  Cal_2_6_5    4.03
 4 lc_4      5    11 41.8      0 cal   Cal_2_5_4  Cal_2_6_5    3.73
 5 lc_4      5    14 64.8      0 cal   Cal_2_5_4  Cal_2_6_5    4.17
 6 lc_4      5    19 24.6      0 cal   Cal_2_5_4  Cal_2_6_5    3.20
 7 lc_5      8    12 11.8      0 cal   Cal_2_5_4  Cal_2_6_5    2.47
 8 lc_6      4    16 24.1      0 f1    Lon_1_7_1  Cal_2_5_4    3.18
 9 lc_6      7    14  8.59     0 cal   Cal_2_5_4  Cal_2_6_5    2.15
10 lc_6      8     3  7.80     0 f2    B12_1      B12_11       2.05
11 lc_6      9     5 23.3      0 f1    Cal_2_5_4  Lon_1_10_2   3.15
12 lc_6      9    14 16.1      0 f1    Cal_2_5_4  Lon_1_10_2   2.78
13 lc_6      9    15 16.3      0 f1    Cal_2_5_4  Lon_1_10_2   2.79
14 lc_6      9    17 29.4      0 f1    Cal_2_5_4  Lon_1_10_2   3.38
15 lc_6      9    18 30.8      0 f1    Cal_2_5_4  Lon_1_10_2   3.43
```

It looks like they are just one off screw ups for some reason, so I'm just going to remove them.

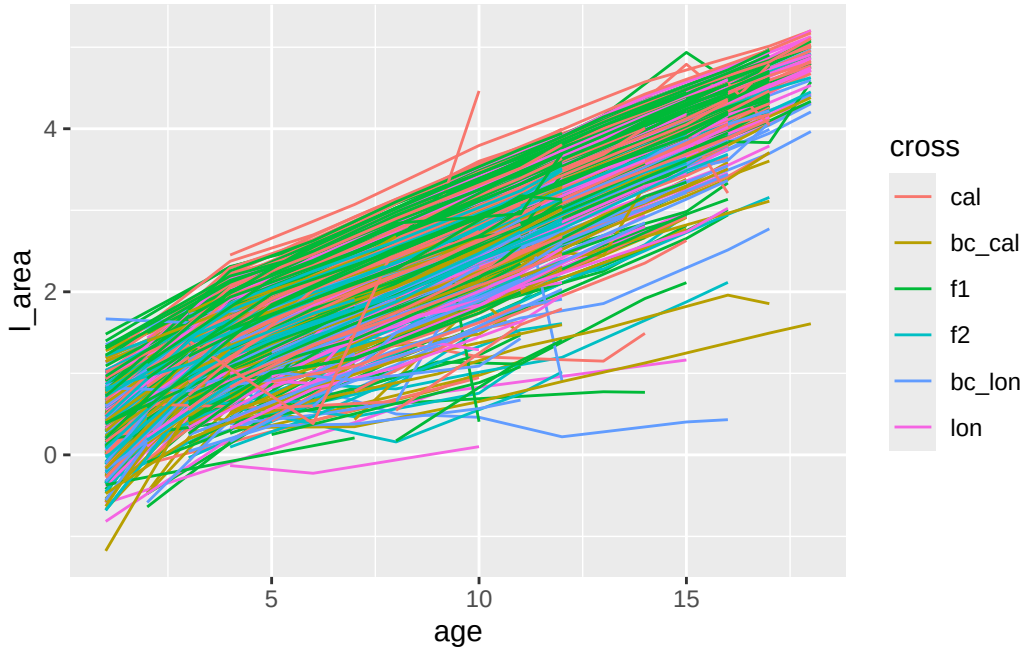
```
seedlings <- seedlings %>%
  mutate(l_area = log(area)) %>%
  filter(!(paste0(tray, block, ind) %in% c("lc_2313", "lc_6911",
                                           "lc_2112", "lc_4912",
                                           "lc_5117", "lc_5117"))) %>%
  filter(l_area > -2.5)
```

Just re-plot without the age 0.

```
seedlings %>%
  filter(age > 0) %>%
  ggplot(aes(x = age, y = l_area)) +
  geom_point()
```




```
seedlings %>%
  filter(age > 0) %>%
  mutate(id = paste0(tray, "_", block, "_", ind)) %>%
  ggplot(aes(x = age, y = l_area, group = id, color = cross)) +
  geom_line()
```



It's hard to make anything from that. Let's just estimate the relative growth rate for each individual.

```
# define a function for doing the linear regression and extracting
# the slope coefficient.

rgr <- function(size, age){
  mod <- lm(size ~ age)
  return(coef(mod)[[2]])
}


(rgr <- seedlings %>%
  filter(age > 0) %>%
  mutate(id = paste0(tray, "-", block, "-", ind)) %>%
  group_by(id) %>%
  summarise(rgr = rgr(l_area, age)) %>%
  separate(id, into = c("tray", "block", "ind"), sep = "-") %>%
  mutate(block = as.integer(block)) %>%
  left_join(ids, by = join_by(tray, block)) %>%
  mutate(cross = tolower(cross)))
```

```
# A tibble: 885 x 7
```

```
   tray  block ind     rgr cross mother father
   <chr> <int> <chr> <dbl> <chr> <chr>  <chr>
 1 lc_1      1 1     0.240 f2    B12_1  B12_11
 2 lc_1      1 10    0.253 f2    B12_1  B12_11
 3 lc_1      1 11    0.186 f2    B12_1  B12_11
 4 lc_1      1 13    0.215 f2    B12_1  B12_11
 5 lc_1      1 14    0.249 f2    B12_1  B12_11
 6 lc_1      1 2     0.206 f2    B12_1  B12_11
 7 lc_1      1 3     0.254 f2    B12_1  B12_11
 8 lc_1      1 4     0.204 f2    B12_1  B12_11
 9 lc_1      1 5     0.206 f2    B12_1  B12_11
10 lc_1      1 7     0.207 f2    B12_1  B12_11
# i 875 more rows
```

Now lets plot the RGRs by cross.

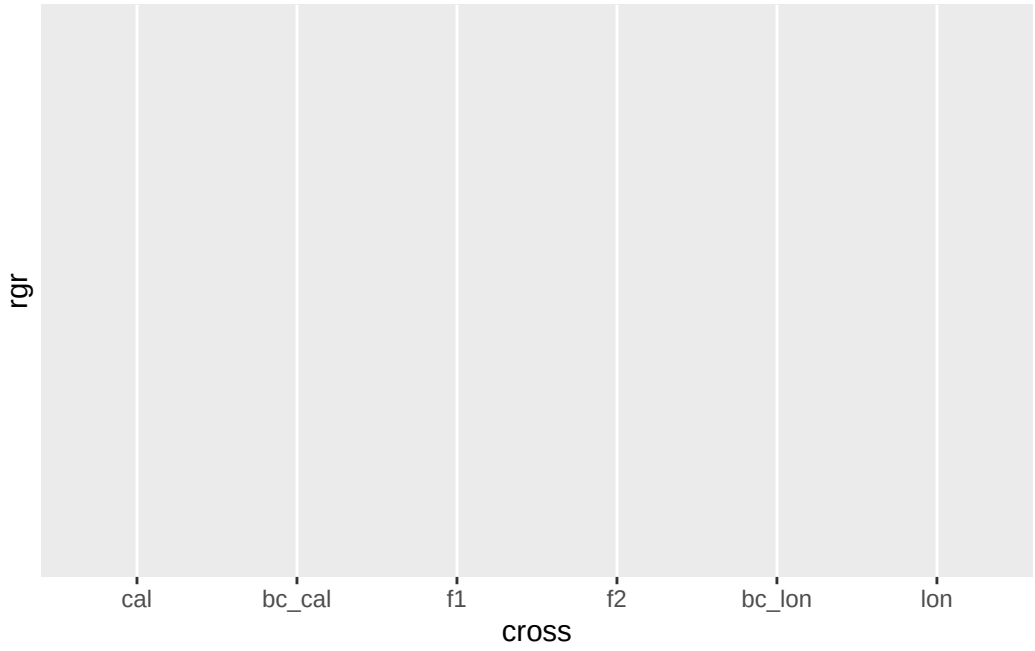
```
rgr %>%
  mutate(cross = factor(cross, levels = c("cal", "bc_cal", "f1", "f2", "bc_lon", "lon"))) %>%
  ggplot(aes(x = cross, y = rgr)) +
  stat_summary(fun.data = mean_cl_boot, geom = "point") +
  stat_summary(fun.data = mean_cl_boot, geom = "errorbar")
```

```
Warning: Removed 7 rows containing non-finite outside the scale range
(`stat_summary()`).
```

```
Warning: Computation failed in `stat_summary()`.
Caused by error in `fun.data()`:
! The package "Hmisc" is required.
```

```
Warning: Removed 7 rows containing non-finite outside the scale range
(`stat_summary()`).
```

```
Warning: Computation failed in `stat_summary()`.
Caused by error in `fun.data()`:
! The package "Hmisc" is required.
```



Hard to see much differences here, but will wait for the full model to see.

Survival Data

Finally, I need to get the survival data in shape.

```
survival <- read.csv(here("data/raw/terminal_drought_trial.csv"))
```

```
(survival <- survival %>%
  mutate(ind = id) %>%
  select(tray, block, ind, alive_nov_9) %>%
  left_join(germ_day) %>%
  drop_na(germ_day) %>%
  mutate(alive_nov_9 = ifelse(is.na(alive_nov_9), 0, 1)))
```

```
Joining with `by = join_by(tray, block, ind)`
```

```
    tray block ind alive_nov_9 germ_day  cross     mother     father
1   lc_2     1   1           0      265 bc_lon      B12_1 Lon_1_10_2
2   lc_2     1   3           0      264 bc_lon      B12_1 Lon_1_10_2
3   lc_2     1   4           0      265 bc_lon      B12_1 Lon_1_10_2
```

```
4   lc_2     1   6          0      264 bc_lon      B12_1 Lon_1_10_2
5   lc_2     1   7          0      277 bc_lon      B12_1 Lon_1_10_2
6   lc_2     1   9          0      263 bc_lon      B12_1 Lon_1_10_2
7   lc_2     1  10          0      270 bc_lon      B12_1 Lon_1_10_2
8   lc_2     1  12          0      266 bc_lon      B12_1 Lon_1_10_2
9   lc_2     1  13          0      265 bc_lon      B12_1 Lon_1_10_2
10  lc_2     1  14          0      267 bc_lon      B12_1 Lon_1_10_2
11  lc_2     1  15          0      265 bc_lon      B12_1 Lon_1_10_2
12  lc_2     1  16          0      269 bc_lon      B12_1 Lon_1_10_2
13  lc_2     1  17          0      269 bc_lon      B12_1 Lon_1_10_2
14  lc_2     1  18          0      265 bc_lon      B12_1 Lon_1_10_2
15  lc_2     2   1          0      270 bc_lon      B12_1 Lon_1_10_2
16  lc_2     2   2          0      270 bc_lon      B12_1 Lon_1_10_2
17  lc_2     2   5          0      274 bc_lon      B12_1 Lon_1_10_2
18  lc_2     2   6          0      267 bc_lon      B12_1 Lon_1_10_2
19  lc_2     2   7          0      265 bc_lon      B12_1 Lon_1_10_2
20  lc_2     2   8          0      263 bc_lon      B12_1 Lon_1_10_2
21  lc_2     2   9          0      269 bc_lon      B12_1 Lon_1_10_2
22  lc_2     2  10          0      266 bc_lon      B12_1 Lon_1_10_2
23  lc_2     2  12          0      277 bc_lon      B12_1 Lon_1_10_2
24  lc_2     2  13          0      269 bc_lon      B12_1 Lon_1_10_2
25  lc_2     2  14          1      263 bc_lon      B12_1 Lon_1_10_2
26  lc_2     2  15          0      277 bc_lon      B12_1 Lon_1_10_2
27  lc_2     2  16          0      270 bc_lon      B12_1 Lon_1_10_2
28  lc_2     2  20          0      270 bc_lon      B12_1 Lon_1_10_2
29  lc_2     3   1          0      269      f1  Lon_1_7_1  Cal_2_5_4
30  lc_2     3   2          0      266      f1  Lon_1_7_1  Cal_2_5_4
31  lc_2     3   3          0      269      f1  Lon_1_7_1  Cal_2_5_4
32  lc_2     3   4          0      265      f1  Lon_1_7_1  Cal_2_5_4
33  lc_2     3   5          0      270      f1  Lon_1_7_1  Cal_2_5_4
34  lc_2     3   6          0      269      f1  Lon_1_7_1  Cal_2_5_4
35  lc_2     3   7          1      264      f1  Lon_1_7_1  Cal_2_5_4
36  lc_2     3   8          1      264      f1  Lon_1_7_1  Cal_2_5_4
37  lc_2     3   9          0      265      f1  Lon_1_7_1  Cal_2_5_4
38  lc_2     3  11          1      263      f1  Lon_1_7_1  Cal_2_5_4
39  lc_2     3  12          1      265      f1  Lon_1_7_1  Cal_2_5_4
40  lc_2     3  13          0      277      f1  Lon_1_7_1  Cal_2_5_4
41  lc_2     3  14          0      267      f1  Lon_1_7_1  Cal_2_5_4
42  lc_2     3  15          0      263      f1  Lon_1_7_1  Cal_2_5_4
43  lc_2     3  16          0      269      f1  Lon_1_7_1  Cal_2_5_4
44  lc_2     3  17          0      266      f1  Lon_1_7_1  Cal_2_5_4
45  lc_2     3  18          0      269      f1  Lon_1_7_1  Cal_2_5_4
46  lc_2     4   1          0      265     cal  Cal_2_5_4  Cal_2_6_5
```

47	lc_2	4	2	0	265	cal	Cal_2_5_4	Cal_2_6_5
48	lc_2	4	3	0	269	cal	Cal_2_5_4	Cal_2_6_5
49	lc_2	4	5	0	269	cal	Cal_2_5_4	Cal_2_6_5
50	lc_2	4	6	0	264	cal	Cal_2_5_4	Cal_2_6_5
51	lc_2	4	9	0	269	cal	Cal_2_5_4	Cal_2_6_5
52	lc_2	4	10	0	263	cal	Cal_2_5_4	Cal_2_6_5
53	lc_2	4	11	1	264	cal	Cal_2_5_4	Cal_2_6_5
54	lc_2	4	12	1	264	cal	Cal_2_5_4	Cal_2_6_5
55	lc_2	4	13	0	263	cal	Cal_2_5_4	Cal_2_6_5
56	lc_2	4	14	0	263	cal	Cal_2_5_4	Cal_2_6_5
57	lc_2	4	16	0	271	cal	Cal_2_5_4	Cal_2_6_5
58	lc_2	4	17	0	263	cal	Cal_2_5_4	Cal_2_6_5
59	lc_2	4	18	0	270	cal	Cal_2_5_4	Cal_2_6_5
60	lc_2	4	19	0	265	cal	Cal_2_5_4	Cal_2_6_5
61	lc_2	4	20	0	263	cal	Cal_2_5_4	Cal_2_6_5
62	lc_2	5	1	0	266	f1	Cal_2_5_4	Lon_1_7_1
63	lc_2	5	3	0	269	f1	Cal_2_5_4	Lon_1_7_1
64	lc_2	5	5	0	270	f1	Cal_2_5_4	Lon_1_7_1
65	lc_2	5	6	0	266	f1	Cal_2_5_4	Lon_1_7_1
66	lc_2	5	7	0	265	f1	Cal_2_5_4	Lon_1_7_1
67	lc_2	5	8	1	264	f1	Cal_2_5_4	Lon_1_7_1
68	lc_2	5	9	0	265	f1	Cal_2_5_4	Lon_1_7_1
69	lc_2	5	10	1	264	f1	Cal_2_5_4	Lon_1_7_1
70	lc_2	5	11	1	265	f1	Cal_2_5_4	Lon_1_7_1
71	lc_2	5	12	1	264	f1	Cal_2_5_4	Lon_1_7_1
72	lc_2	5	13	0	266	f1	Cal_2_5_4	Lon_1_7_1
73	lc_2	5	14	1	263	f1	Cal_2_5_4	Lon_1_7_1
74	lc_2	5	15	0	265	f1	Cal_2_5_4	Lon_1_7_1
75	lc_2	5	16	1	265	f1	Cal_2_5_4	Lon_1_7_1
76	lc_2	5	18	0	263	f1	Cal_2_5_4	Lon_1_7_1
77	lc_2	6	2	1	263	cal	Cal_2_6_5	Cal_2_5_4
78	lc_2	6	3	0	269	cal	Cal_2_6_5	Cal_2_5_4
79	lc_2	6	4	1	269	cal	Cal_2_6_5	Cal_2_5_4
80	lc_2	6	5	0	269	cal	Cal_2_6_5	Cal_2_5_4
81	lc_2	6	6	0	274	cal	Cal_2_6_5	Cal_2_5_4
82	lc_2	6	7	0	269	cal	Cal_2_6_5	Cal_2_5_4
83	lc_2	6	8	1	264	cal	Cal_2_6_5	Cal_2_5_4
84	lc_2	6	9	0	265	cal	Cal_2_6_5	Cal_2_5_4
85	lc_2	6	10	0	269	cal	Cal_2_6_5	Cal_2_5_4
86	lc_2	6	11	1	265	cal	Cal_2_6_5	Cal_2_5_4
87	lc_2	6	12	0	269	cal	Cal_2_6_5	Cal_2_5_4
88	lc_2	6	13	1	264	cal	Cal_2_6_5	Cal_2_5_4
89	lc_2	6	14	0	265	cal	Cal_2_6_5	Cal_2_5_4

```
90  lc_2     6  15          0      269    cal  Cal_2_6_5  Cal_2_5_4
91  lc_2     6  16          1      266    cal  Cal_2_6_5  Cal_2_5_4
92  lc_2     6  17          1      263    cal  Cal_2_6_5  Cal_2_5_4
93  lc_2     6  18          0      274    cal  Cal_2_6_5  Cal_2_5_4
94  lc_2     6  20          0      269    cal  Cal_2_6_5  Cal_2_5_4
95  lc_2     7   1          1      263    lon Lon_1_10_2  Lon_1_7_1
96  lc_2     7   4          0      266    lon Lon_1_10_2  Lon_1_7_1
97  lc_2     7   5          0      269    lon Lon_1_10_2  Lon_1_7_1
98  lc_2     7   6          1      265    lon Lon_1_10_2  Lon_1_7_1
99  lc_2     7   7          1      266    lon Lon_1_10_2  Lon_1_7_1
100 lc_2     7   8          1      263    lon Lon_1_10_2  Lon_1_7_1
101 lc_2     7   9          1      263    lon Lon_1_10_2  Lon_1_7_1
102 lc_2     7  10          1      263    lon Lon_1_10_2  Lon_1_7_1
103 lc_2     7  11          0      274    lon Lon_1_10_2  Lon_1_7_1
104 lc_2     7  12          1      264    lon Lon_1_10_2  Lon_1_7_1
105 lc_2     7  13          1      263    lon Lon_1_10_2  Lon_1_7_1
106 lc_2     7  14          1      265    lon Lon_1_10_2  Lon_1_7_1
107 lc_2     7  15          1      265    lon Lon_1_10_2  Lon_1_7_1
108 lc_2     7  16          1      264    lon Lon_1_10_2  Lon_1_7_1
109 lc_2     7  17          0      280    lon Lon_1_10_2  Lon_1_7_1
110 lc_2     7  18          0      266    lon Lon_1_10_2  Lon_1_7_1
111 lc_2     7  19          1      263    lon Lon_1_10_2  Lon_1_7_1
112 lc_2     7  20          1      266    lon Lon_1_10_2  Lon_1_7_1
113 lc_2     8   1          1      265     f2      B12_1     B12_11
114 lc_2     8   2          0      265     f2      B12_1     B12_11
115 lc_2     8   3          0      269     f2      B12_1     B12_11
116 lc_2     8   4          1      263     f2      B12_1     B12_11
117 lc_2     8   5          0      265     f2      B12_1     B12_11
118 lc_2     8   6          0      273     f2      B12_1     B12_11
119 lc_2     8   7          0      274     f2      B12_1     B12_11
120 lc_2     8   8          0      269     f2      B12_1     B12_11
121 lc_2     8   9          0      269     f2      B12_1     B12_11
122 lc_2     8  10          0      270     f2      B12_1     B12_11
123 lc_2     8  11          0      269     f2      B12_1     B12_11
124 lc_2     8  12          0      270     f2      B12_1     B12_11
125 lc_2     8  13          0      266     f2      B12_1     B12_11
126 lc_2     8  15          0      269     f2      B12_1     B12_11
127 lc_2     8  16          0      269     f2      B12_1     B12_11
128 lc_2     8  18          0      270     f2      B12_1     B12_11
129 lc_2     8  20          0      266     f2      B12_1     B12_11
130 lc_2     9   1          0      266 bc_cal      B12_1  Cal_2_5_4
131 lc_2     9   2          0      269 bc_cal      B12_1  Cal_2_5_4
132 lc_2     9   3          0      269 bc_cal      B12_1  Cal_2_5_4
```

```
133 lc_2     9   4           0      269 bc_cal      B12_1  Cal_2_5_4
134 lc_2     9   5           1      265 bc_cal      B12_1  Cal_2_5_4
135 lc_2     9   6           0      265 bc_cal      B12_1  Cal_2_5_4
136 lc_2     9   8           0      269 bc_cal      B12_1  Cal_2_5_4
137 lc_2     9   9           0      269 bc_cal      B12_1  Cal_2_5_4
138 lc_2     9  10           0      269 bc_cal      B12_1  Cal_2_5_4
139 lc_2     9  11           0      270 bc_cal      B12_1  Cal_2_5_4
140 lc_2     9  12           0      265 bc_cal      B12_1  Cal_2_5_4
141 lc_2     9  13           0      269 bc_cal      B12_1  Cal_2_5_4
142 lc_2     9  14           0      267 bc_cal      B12_1  Cal_2_5_4
143 lc_2     9  15           0      271 bc_cal      B12_1  Cal_2_5_4
144 lc_2     9  16           0      269 bc_cal      B12_1  Cal_2_5_4
145 lc_2     9  17           0      263 bc_cal      B12_1  Cal_2_5_4
146 lc_2     9  18           0      269 bc_cal      B12_1  Cal_2_5_4
147 lc_2     9  19           0      264 bc_cal      B12_1  Cal_2_5_4
148 lc_2    10   1           0      264    lon  Lon_1_7_1 Lon_1_10_2
149 lc_2    10   2           0      267    lon  Lon_1_7_1 Lon_1_10_2
150 lc_2    10   3           0      265    lon  Lon_1_7_1 Lon_1_10_2
151 lc_2    10   5           1      265    lon  Lon_1_7_1 Lon_1_10_2
152 lc_2    10   7           0      269    lon  Lon_1_7_1 Lon_1_10_2
153 lc_2    10   9           0      271    lon  Lon_1_7_1 Lon_1_10_2
154 lc_2    10  10           0      267    lon  Lon_1_7_1 Lon_1_10_2
155 lc_2    10  11           1      264    lon  Lon_1_7_1 Lon_1_10_2
156 lc_2    10  14           1      265    lon  Lon_1_7_1 Lon_1_10_2
157 lc_2    10  16           1      264    lon  Lon_1_7_1 Lon_1_10_2
158 lc_2    10  17           0      265    lon  Lon_1_7_1 Lon_1_10_2
159 lc_2    10  18           0      267    lon  Lon_1_7_1 Lon_1_10_2
160 lc_2    10  19           1      265    lon  Lon_1_7_1 Lon_1_10_2
161 lc_2    11   1           0      269     f2      B12_1     B12_11
162 lc_2    11   3           0      265     f2      B12_1     B12_11
163 lc_2    11   4           0      270     f2      B12_1     B12_11
164 lc_2    11   5           0      270     f2      B12_1     B12_11
165 lc_2    11   8           0      269     f2      B12_1     B12_11
166 lc_2    11   9           0      281     f2      B12_1     B12_11
167 lc_2    11  11           0      269     f2      B12_1     B12_11
168 lc_2    11  12           0      269     f2      B12_1     B12_11
169 lc_2    11  13           1      266     f2      B12_1     B12_11
170 lc_2    11  15           0      266     f2      B12_1     B12_11
171 lc_2    11  16           0      269     f2      B12_1     B12_11
172 lc_2    11  17           0      277     f2      B12_1     B12_11
173 lc_2    11  18           1      269     f2      B12_1     B12_11
174 lc_2    11  19           0      265     f2      B12_1     B12_11
175 lc_2    11  20           0      274     f2      B12_1     B12_11
```

```
176 lc_2    12   1           0      265 bc_cal      B12_1  Cal_2_5_4
177 lc_2    12   2           1      265 bc_cal      B12_1  Cal_2_5_4
178 lc_2    12   3           0      267 bc_cal      B12_1  Cal_2_5_4
179 lc_2    12   4           1      263 bc_cal      B12_1  Cal_2_5_4
180 lc_2    12   5           0      269 bc_cal      B12_1  Cal_2_5_4
181 lc_2    12   6           1      264 bc_cal      B12_1  Cal_2_5_4
182 lc_2    12   7           0      274 bc_cal      B12_1  Cal_2_5_4
183 lc_2    12   8           0      265 bc_cal      B12_1  Cal_2_5_4
184 lc_2    12  10           0      269 bc_cal      B12_1  Cal_2_5_4
185 lc_2    12  11           0      270 bc_cal      B12_1  Cal_2_5_4
186 lc_2    12  12           0      267 bc_cal      B12_1  Cal_2_5_4
187 lc_2    12  13           0      274 bc_cal      B12_1  Cal_2_5_4
188 lc_2    12  14           0      267 bc_cal      B12_1  Cal_2_5_4
189 lc_2    12  15           0      270 bc_cal      B12_1  Cal_2_5_4
190 lc_2    12  16           0      266 bc_cal      B12_1  Cal_2_5_4
191 lc_2    12  17           0      267 bc_cal      B12_1  Cal_2_5_4
192 lc_2    12  18           0      270 bc_cal      B12_1  Cal_2_5_4
193 lc_2    12  19           0      267 bc_cal      B12_1  Cal_2_5_4
194 lc_2    12  20           0      264 bc_cal      B12_1  Cal_2_5_4
195 lc_4     1   3           0      269    lon  Lon_1_7_1 Lon_1_10_2
196 lc_4     1  11           0      270    lon  Lon_1_7_1 Lon_1_10_2
197 lc_4     1  14           0      264    lon  Lon_1_7_1 Lon_1_10_2
198 lc_4     1  15           0      270    lon  Lon_1_7_1 Lon_1_10_2
199 lc_4     1  17           0      266    lon  Lon_1_7_1 Lon_1_10_2
200 lc_4     2   1           0      266     f1  Cal_2_5_4  Lon_1_7_1
201 lc_4     2   2           1      264     f1  Cal_2_5_4  Lon_1_7_1
202 lc_4     2   3           1      264     f1  Cal_2_5_4  Lon_1_7_1
203 lc_4     2   4           1      269     f1  Cal_2_5_4  Lon_1_7_1
204 lc_4     2   5           0      269     f1  Cal_2_5_4  Lon_1_7_1
205 lc_4     2   6           0      264     f1  Cal_2_5_4  Lon_1_7_1
206 lc_4     2   7           0      267     f1  Cal_2_5_4  Lon_1_7_1
207 lc_4     2   8           0      269     f1  Cal_2_5_4  Lon_1_7_1
208 lc_4     2  10           1      265     f1  Cal_2_5_4  Lon_1_7_1
209 lc_4     2  13           0      269     f1  Cal_2_5_4  Lon_1_7_1
210 lc_4     2  16           0      266     f1  Cal_2_5_4  Lon_1_7_1
211 lc_4     3   1           0      269     f2      B12_1     B12_11
212 lc_4     3   2           0      269     f2      B12_1     B12_11
213 lc_4     3   5           0      269     f2      B12_1     B12_11
214 lc_4     3   6           0      269     f2      B12_1     B12_11
215 lc_4     3   7           1      269     f2      B12_1     B12_11
216 lc_4     3   8           0      269     f2      B12_1     B12_11
217 lc_4     3   9           0      269     f2      B12_1     B12_11
218 lc_4     3  10           0      269     f2      B12_1     B12_11
```

219	lc_4	3	11	0	273	f2	B12_1	B12_11
220	lc_4	3	12	0	269	f2	B12_1	B12_11
221	lc_4	3	13	0	269	f2	B12_1	B12_11
222	lc_4	3	14	0	269	f2	B12_1	B12_11
223	lc_4	3	15	0	280	f2	B12_1	B12_11
224	lc_4	3	16	0	270	f2	B12_1	B12_11
225	lc_4	3	17	0	269	f2	B12_1	B12_11
226	lc_4	3	19	1	266	f2	B12_1	B12_11
227	lc_4	4	1	0	270	bc_lon	B12_1	Lon_1_10_2
228	lc_4	4	2	0	266	bc_lon	B12_1	Lon_1_10_2
229	lc_4	4	3	0	269	bc_lon	B12_1	Lon_1_10_2
230	lc_4	4	7	0	269	bc_lon	B12_1	Lon_1_10_2
231	lc_4	4	9	0	271	bc_lon	B12_1	Lon_1_10_2
232	lc_4	4	13	0	265	bc_lon	B12_1	Lon_1_10_2
233	lc_4	4	14	0	269	bc_lon	B12_1	Lon_1_10_2
234	lc_4	4	16	0	269	bc_lon	B12_1	Lon_1_10_2
235	lc_4	4	19	0	269	bc_lon	B12_1	Lon_1_10_2
236	lc_4	4	20	0	269	bc_lon	B12_1	Lon_1_10_2
237	lc_4	5	8	0	264	cal	Cal_2_5_4	Cal_2_6_5
238	lc_4	5	9	0	281	cal	Cal_2_5_4	Cal_2_6_5
239	lc_4	5	10	0	267	cal	Cal_2_5_4	Cal_2_6_5
240	lc_4	5	11	0	281	cal	Cal_2_5_4	Cal_2_6_5
241	lc_4	5	12	0	265	cal	Cal_2_5_4	Cal_2_6_5
242	lc_4	5	13	1	265	cal	Cal_2_5_4	Cal_2_6_5
243	lc_4	5	14	0	281	cal	Cal_2_5_4	Cal_2_6_5
244	lc_4	5	15	1	265	cal	Cal_2_5_4	Cal_2_6_5
245	lc_4	5	16	1	264	cal	Cal_2_5_4	Cal_2_6_5
246	lc_4	5	17	1	264	cal	Cal_2_5_4	Cal_2_6_5
247	lc_4	5	18	0	271	cal	Cal_2_5_4	Cal_2_6_5
248	lc_4	5	19	0	281	cal	Cal_2_5_4	Cal_2_6_5
249	lc_4	6	1	0	269	bc_cal	B12_1	Cal_2_5_4
250	lc_4	6	2	0	269	bc_cal	B12_1	Cal_2_5_4
251	lc_4	6	3	0	269	bc_cal	B12_1	Cal_2_5_4
252	lc_4	6	4	1	267	bc_cal	B12_1	Cal_2_5_4
253	lc_4	6	5	0	269	bc_cal	B12_1	Cal_2_5_4
254	lc_4	6	7	0	273	bc_cal	B12_1	Cal_2_5_4
255	lc_4	6	9	1	264	bc_cal	B12_1	Cal_2_5_4
256	lc_4	6	11	0	274	bc_cal	B12_1	Cal_2_5_4
257	lc_4	6	13	1	266	bc_cal	B12_1	Cal_2_5_4
258	lc_4	6	14	0	264	bc_cal	B12_1	Cal_2_5_4
259	lc_4	7	7	0	270	bc_lon	B12_1	Lon_1_10_2
260	lc_4	7	8	0	274	bc_lon	B12_1	Lon_1_10_2
261	lc_4	7	9	0	277	bc_lon	B12_1	Lon_1_10_2

```
262 lc_4     7  12          1      270 bc_lon      B12_1 Lon_1_10_2
263 lc_4     7  13          0      271 bc_lon      B12_1 Lon_1_10_2
264 lc_4     7  15          1      269 bc_lon      B12_1 Lon_1_10_2
265 lc_4     7  17          0      269 bc_lon      B12_1 Lon_1_10_2
266 lc_4     7  18          0      270 bc_lon      B12_1 Lon_1_10_2
267 lc_4     8   1          0      274     f1  Lon_1_7_1  Cal_2_5_4
268 lc_4     8   2          0      271     f1  Lon_1_7_1  Cal_2_5_4
269 lc_4     8   3          1      266     f1  Lon_1_7_1  Cal_2_5_4
270 lc_4     8   5          1      266     f1  Lon_1_7_1  Cal_2_5_4
271 lc_4     8   6          0      265     f1  Lon_1_7_1  Cal_2_5_4
272 lc_4     8   8          1      265     f1  Lon_1_7_1  Cal_2_5_4
273 lc_4     8   9          0      269     f1  Lon_1_7_1  Cal_2_5_4
274 lc_4     8  10          0      269     f1  Lon_1_7_1  Cal_2_5_4
275 lc_4     8  11          0      274     f1  Lon_1_7_1  Cal_2_5_4
276 lc_4     8  12          0      266     f1  Lon_1_7_1  Cal_2_5_4
277 lc_4     8  15          0      269     f1  Lon_1_7_1  Cal_2_5_4
278 lc_4     8  17          0      265     f1  Lon_1_7_1  Cal_2_5_4
279 lc_4     9   1          0      270    cal  Cal_2_6_5  Cal_2_5_4
280 lc_4     9   2          0      271    cal  Cal_2_6_5  Cal_2_5_4
281 lc_4     9   4          0      269    cal  Cal_2_6_5  Cal_2_5_4
282 lc_4     9   5          0      269    cal  Cal_2_6_5  Cal_2_5_4
283 lc_4     9   6          0      281    cal  Cal_2_6_5  Cal_2_5_4
284 lc_4     9   7          0      266    cal  Cal_2_6_5  Cal_2_5_4
285 lc_4     9   9          0      281    cal  Cal_2_6_5  Cal_2_5_4
286 lc_4     9  11          0      274    cal  Cal_2_6_5  Cal_2_5_4
287 lc_4     9  12          0      263    cal  Cal_2_6_5  Cal_2_5_4
288 lc_4     9  17          0      269    cal  Cal_2_6_5  Cal_2_5_4
289 lc_4     9  18          1      266    cal  Cal_2_6_5  Cal_2_5_4
290 lc_4    10   2          0      269 bc_cal      B12_1  Cal_2_5_4
291 lc_4    10   5          0      263 bc_cal      B12_1  Cal_2_5_4
292 lc_4    10   7          1      269 bc_cal      B12_1  Cal_2_5_4
293 lc_4    10   8          0      274 bc_cal      B12_1  Cal_2_5_4
294 lc_4    10  11          0      269 bc_cal      B12_1  Cal_2_5_4
295 lc_4    10  12          0      264 bc_cal      B12_1  Cal_2_5_4
296 lc_4    10  15          0      277 bc_cal      B12_1  Cal_2_5_4
297 lc_4    10  16          0      266 bc_cal      B12_1  Cal_2_5_4
298 lc_4    10  17          1      269 bc_cal      B12_1  Cal_2_5_4
299 lc_4    10  18          1      271 bc_cal      B12_1  Cal_2_5_4
300 lc_4    11   2          1      270     f2      B12_1     B12_11
301 lc_4    11   3          0      269     f2      B12_1     B12_11
302 lc_4    11   4          0      281     f2      B12_1     B12_11
303 lc_4    11   6          1      270     f2      B12_1     B12_11
304 lc_4    11   7          0      270     f2      B12_1     B12_11
```

305	lc_4	11	8	0	274	f2	B12_1	B12_11
306	lc_4	11	9	0	269	f2	B12_1	B12_11
307	lc_4	12	1	1	269	lon	Lon_1_10_2	Lon_1_7_1
308	lc_4	12	2	1	266	lon	Lon_1_10_2	Lon_1_7_1
309	lc_4	12	3	0	271	lon	Lon_1_10_2	Lon_1_7_1
310	lc_4	12	5	1	265	lon	Lon_1_10_2	Lon_1_7_1
311	lc_4	12	7	1	266	lon	Lon_1_10_2	Lon_1_7_1
312	lc_4	12	10	0	270	lon	Lon_1_10_2	Lon_1_7_1
313	lc_4	12	13	1	264	lon	Lon_1_10_2	Lon_1_7_1
314	lc_4	12	14	0	265	lon	Lon_1_10_2	Lon_1_7_1
315	lc_4	12	15	0	277	lon	Lon_1_10_2	Lon_1_7_1
316	lc_4	12	17	0	270	lon	Lon_1_10_2	Lon_1_7_1
317	lc_4	12	18	0	265	lon	Lon_1_10_2	Lon_1_7_1
318	lc_4	12	19	1	263	lon	Lon_1_10_2	Lon_1_7_1
319	lc_4	12	20	0	274	lon	Lon_1_10_2	Lon_1_7_1
320	lc_5	1	4	0	267	cal	Cal_2_5_4	Cal_2_6_5
321	lc_5	1	6	1	264	cal	Cal_2_5_4	Cal_2_6_5
322	lc_5	1	8	0	266	cal	Cal_2_5_4	Cal_2_6_5
323	lc_5	1	9	0	266	cal	Cal_2_5_4	Cal_2_6_5
324	lc_5	1	10	0	270	cal	Cal_2_5_4	Cal_2_6_5
325	lc_5	1	15	1	264	cal	Cal_2_5_4	Cal_2_6_5
326	lc_5	1	16	0	270	cal	Cal_2_5_4	Cal_2_6_5
327	lc_5	1	17	0	265	cal	Cal_2_5_4	Cal_2_6_5
328	lc_5	2	3	0	269	lon	Lon_1_7_1	Lon_1_10_2
329	lc_5	2	6	1	265	lon	Lon_1_7_1	Lon_1_10_2
330	lc_5	2	7	1	265	lon	Lon_1_7_1	Lon_1_10_2
331	lc_5	2	8	0	271	lon	Lon_1_7_1	Lon_1_10_2
332	lc_5	2	9	0	269	lon	Lon_1_7_1	Lon_1_10_2
333	lc_5	2	14	0	269	lon	Lon_1_7_1	Lon_1_10_2
334	lc_5	2	15	1	265	lon	Lon_1_7_1	Lon_1_10_2
335	lc_5	2	17	0	269	lon	Lon_1_7_1	Lon_1_10_2
336	lc_5	3	2	0	269	bc_lon	B12_1	Lon_1_10_2
337	lc_5	3	5	0	269	bc_lon	B12_1	Lon_1_10_2
338	lc_5	3	6	0	269	bc_lon	B12_1	Lon_1_10_2
339	lc_5	3	8	0	265	bc_lon	B12_1	Lon_1_10_2
340	lc_5	3	11	0	277	bc_lon	B12_1	Lon_1_10_2
341	lc_5	3	12	0	274	bc_lon	B12_1	Lon_1_10_2
342	lc_5	3	13	1	266	bc_lon	B12_1	Lon_1_10_2
343	lc_5	3	15	0	270	bc_lon	B12_1	Lon_1_10_2
344	lc_5	3	16	0	274	bc_lon	B12_1	Lon_1_10_2
345	lc_5	3	17	0	274	bc_lon	B12_1	Lon_1_10_2
346	lc_5	3	19	1	266	bc_lon	B12_1	Lon_1_10_2
347	lc_5	3	20	0	273	bc_lon	B12_1	Lon_1_10_2

348	lc_5	4	2	0	270	f2	B12_1	B12_11
349	lc_5	4	3	0	270	f2	B12_1	B12_11
350	lc_5	4	4	0	277	f2	B12_1	B12_11
351	lc_5	4	6	1	263	f2	B12_1	B12_11
352	lc_5	4	8	0	270	f2	B12_1	B12_11
353	lc_5	4	9	0	267	f2	B12_1	B12_11
354	lc_5	4	12	1	267	f2	B12_1	B12_11
355	lc_5	4	14	0	266	f2	B12_1	B12_11
356	lc_5	4	15	0	269	f2	B12_1	B12_11
357	lc_5	4	17	0	266	f2	B12_1	B12_11
358	lc_5	4	18	0	270	f2	B12_1	B12_11
359	lc_5	4	19	0	269	f2	B12_1	B12_11
360	lc_5	5	1	1	264	f1	Lon_1_10_2	Cal_2_5_4
361	lc_5	5	4	0	269	f1	Lon_1_10_2	Cal_2_5_4
362	lc_5	5	7	1	269	f1	Lon_1_10_2	Cal_2_5_4
363	lc_5	5	9	0	269	f1	Lon_1_10_2	Cal_2_5_4
364	lc_5	5	12	1	265	f1	Lon_1_10_2	Cal_2_5_4
365	lc_5	5	13	1	265	f1	Lon_1_10_2	Cal_2_5_4
366	lc_5	5	15	0	269	f1	Lon_1_10_2	Cal_2_5_4
367	lc_5	5	16	1	265	f1	Lon_1_10_2	Cal_2_5_4
368	lc_5	5	18	0	269	f1	Lon_1_10_2	Cal_2_5_4
369	lc_5	5	19	0	267	f1	Lon_1_10_2	Cal_2_5_4
370	lc_5	6	1	0	270	f2	B12_1	B12_11
371	lc_5	6	2	0	269	f2	B12_1	B12_11
372	lc_5	6	3	0	270	f2	B12_1	B12_11
373	lc_5	6	4	0	269	f2	B12_1	B12_11
374	lc_5	6	5	0	270	f2	B12_1	B12_11
375	lc_5	6	6	0	269	f2	B12_1	B12_11
376	lc_5	6	7	0	274	f2	B12_1	B12_11
377	lc_5	6	8	0	277	f2	B12_1	B12_11
378	lc_5	6	9	0	270	f2	B12_1	B12_11
379	lc_5	6	10	0	269	f2	B12_1	B12_11
380	lc_5	6	18	0	269	f2	B12_1	B12_11
381	lc_5	7	2	1	269	f1	Cal_2_5_4	Lon_1_7_1
382	lc_5	7	4	1	266	f1	Cal_2_5_4	Lon_1_7_1
383	lc_5	7	5	1	266	f1	Cal_2_5_4	Lon_1_7_1
384	lc_5	7	7	1	264	f1	Cal_2_5_4	Lon_1_7_1
385	lc_5	7	9	1	264	f1	Cal_2_5_4	Lon_1_7_1
386	lc_5	7	10	1	269	f1	Cal_2_5_4	Lon_1_7_1
387	lc_5	7	11	1	264	f1	Cal_2_5_4	Lon_1_7_1
388	lc_5	7	12	1	264	f1	Cal_2_5_4	Lon_1_7_1
389	lc_5	7	13	1	265	f1	Cal_2_5_4	Lon_1_7_1
390	lc_5	7	14	1	267	f1	Cal_2_5_4	Lon_1_7_1

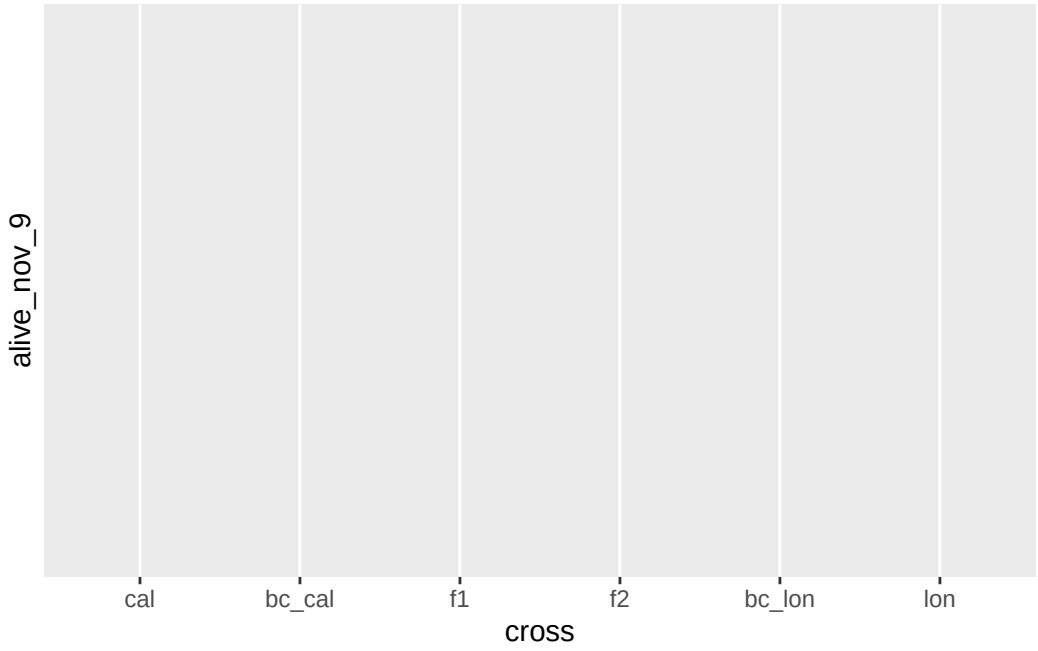
391	lc_5	7	15	1	265	f1	Cal_2_5_4	Lon_1_7_1
392	lc_5	7	16	1	264	f1	Cal_2_5_4	Lon_1_7_1
393	lc_5	7	17	1	264	f1	Cal_2_5_4	Lon_1_7_1
394	lc_5	7	18	1	265	f1	Cal_2_5_4	Lon_1_7_1
395	lc_5	7	19	1	266	f1	Cal_2_5_4	Lon_1_7_1
396	lc_5	7	20	0	274	f1	Cal_2_5_4	Lon_1_7_1
397	lc_5	8	2	0	265	cal	Cal_2_5_4	Cal_2_6_5
398	lc_5	8	3	1	264	cal	Cal_2_5_4	Cal_2_6_5
399	lc_5	8	4	1	269	cal	Cal_2_5_4	Cal_2_6_5
400	lc_5	8	6	1	269	cal	Cal_2_5_4	Cal_2_6_5
401	lc_5	8	7	1	266	cal	Cal_2_5_4	Cal_2_6_5
402	lc_5	8	8	1	264	cal	Cal_2_5_4	Cal_2_6_5
403	lc_5	8	11	0	269	cal	Cal_2_5_4	Cal_2_6_5
404	lc_5	8	12	0	280	cal	Cal_2_5_4	Cal_2_6_5
405	lc_5	8	17	0	265	cal	Cal_2_5_4	Cal_2_6_5
406	lc_5	9	2	0	274	bc_cal	B12_1	Cal_2_5_4
407	lc_5	9	4	0	266	bc_cal	B12_1	Cal_2_5_4
408	lc_5	9	5	0	274	bc_cal	B12_1	Cal_2_5_4
409	lc_5	9	7	0	267	bc_cal	B12_1	Cal_2_5_4
410	lc_5	9	8	0	270	bc_cal	B12_1	Cal_2_5_4
411	lc_5	9	9	0	275	bc_cal	B12_1	Cal_2_5_4
412	lc_5	9	10	0	269	bc_cal	B12_1	Cal_2_5_4
413	lc_5	9	11	0	274	bc_cal	B12_1	Cal_2_5_4
414	lc_5	9	12	0	270	bc_cal	B12_1	Cal_2_5_4
415	lc_5	9	13	0	269	bc_cal	B12_1	Cal_2_5_4
416	lc_5	9	15	0	269	bc_cal	B12_1	Cal_2_5_4
417	lc_5	9	16	0	269	bc_cal	B12_1	Cal_2_5_4
418	lc_5	9	18	0	269	bc_cal	B12_1	Cal_2_5_4
419	lc_5	9	19	0	269	bc_cal	B12_1	Cal_2_5_4
420	lc_5	10	2	0	269	bc_cal	B12_1	Cal_2_5_4
421	lc_5	10	4	0	273	bc_cal	B12_1	Cal_2_5_4
422	lc_5	10	6	0	266	bc_cal	B12_1	Cal_2_5_4
423	lc_5	10	7	0	267	bc_cal	B12_1	Cal_2_5_4
424	lc_5	10	9	1	265	bc_cal	B12_1	Cal_2_5_4
425	lc_5	10	14	0	271	bc_cal	B12_1	Cal_2_5_4
426	lc_5	10	15	0	271	bc_cal	B12_1	Cal_2_5_4
427	lc_5	11	2	0	274	bc_lon	B12_1	Lon_1_10_2
428	lc_5	11	6	0	269	bc_lon	B12_1	Lon_1_10_2
429	lc_5	11	7	0	281	bc_lon	B12_1	Lon_1_10_2
430	lc_5	11	8	0	270	bc_lon	B12_1	Lon_1_10_2
431	lc_5	11	11	1	266	bc_lon	B12_1	Lon_1_10_2
432	lc_5	11	12	0	265	bc_lon	B12_1	Lon_1_10_2
433	lc_5	11	15	1	264	bc_lon	B12_1	Lon_1_10_2

```
434 lc_5    11  16          0      269 bc_lon      B12_1 Lon_1_10_2
435 lc_5    11  17          0      274 bc_lon      B12_1 Lon_1_10_2
436 lc_5    11  20          0      269 bc_lon      B12_1 Lon_1_10_2
437 lc_5    12   1          0      269    lon Lon_1_10_2  Lon_1_7_1
438 lc_5    12   2          1      263    lon Lon_1_10_2  Lon_1_7_1
439 lc_5    12   7          0      266    lon Lon_1_10_2  Lon_1_7_1
440 lc_5    12   8          0      280    lon Lon_1_10_2  Lon_1_7_1
441 lc_5    12  11          0      267    lon Lon_1_10_2  Lon_1_7_1
442 lc_5    12  12          0      266    lon Lon_1_10_2  Lon_1_7_1
443 lc_5    12  15          0      267    lon Lon_1_10_2  Lon_1_7_1
444 lc_5    12  19          0      269    lon Lon_1_10_2  Lon_1_7_1
```

Add the above code to the ‘wrangling.R script.

```
survival %>%
  group_by(cross) %>%
  ggplot(aes(x = cross, y = alive_nov_9)) +
  stat_summary(fun.data = mean_cl_boot, geom = "point") +
  stat_summary(fun.data = mean_cl_boot, geom = "errorbar")
```

```
Warning: Computation failed in `stat_summary()`.
Computation failed in `stat_summary()`.
Caused by error in `fun.data()`:
! The package "Hmisc" is required.
```


alive_nov_9
cal
bc_cal
f1
f2
bc_lon
lon
cross